

4

The Study of Chemical Reactions

Goals for Chapter 4

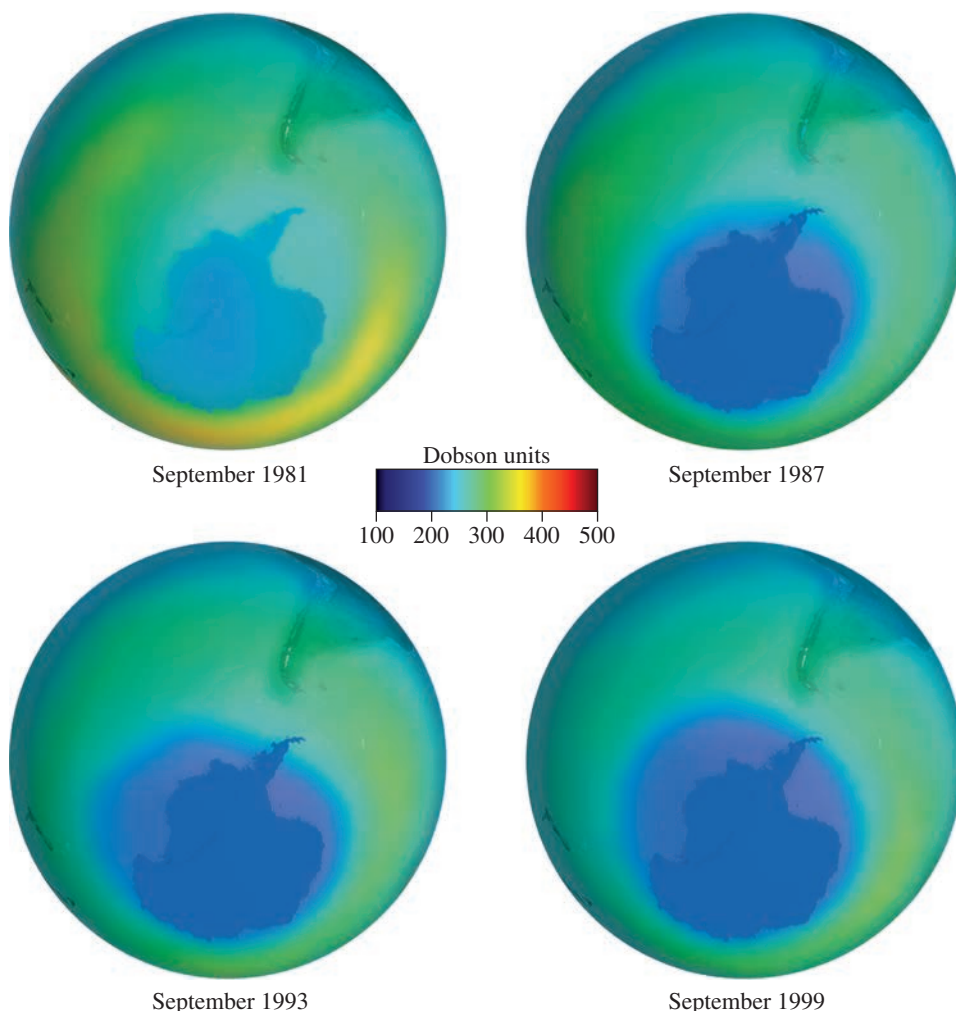
1 Propose mechanisms and explain the steps for simple reactions such as free-radical halogenation.

2 Draw a reaction-energy diagram, and use it to identify the factors controlling the thermodynamics and kinetics of a reaction.

3 Use the mechanism, thermodynamics, and kinetics of a reaction to predict which of several possible products is the major product.

4 Identify reactive intermediates and explain their properties.

◀ This picture shows the “hole” in the protective ozone layer over Antarctica that resulted from the release of chlorofluorocarbons in refrigerants and aerosol spray cans. Chlorofluorocarbon refrigerants such as CF_2Cl_2 (Freon® 12) react with free radicals in the atmosphere to generate chlorine radicals, which catalyze decomposition of the ozone layer. The “hole” is now improving because the worst chlorofluorocarbons are no longer used.



4-1 Introduction

The most interesting part of organic chemistry begins now, with the study of making and breaking bonds. Organic chemists study reactions to learn how to use them to make interesting and useful products. We also want to understand how reactions occur so that we can predict new and useful reactions.

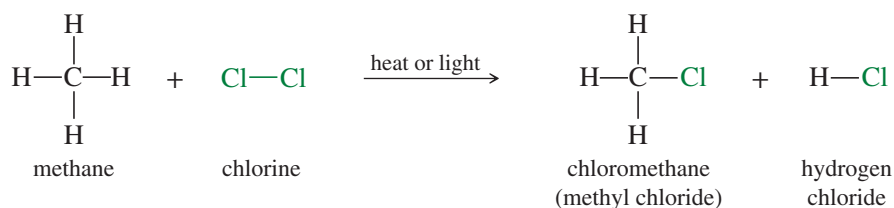
Nobody can remember thousands of specific organic reactions, but we can organize the reactions into logical groups based on how the reactions take place and what intermediates are involved. We begin our study by considering the **halogenation** of alkanes, a relatively simple substitution of a halogen for a hydrogen that can occur in the gas phase, without a solvent to complicate the reaction. In practice, alkanes are so unreactive that they are rarely used as starting materials for laboratory organic syntheses. We start with alkanes, however, because we have already studied their structure and properties, and their reactions are relatively uncomplicated. Once we have used alkanes to introduce the tools for studying reactions, we will apply those tools to a variety of more useful synthetic reactions.

Writing the overall equation, with the reactants on the left and the products on the right, is only the first step in our study of a reaction. If we truly want to understand a reaction, we must also know the *mechanism*, the step-by-step pathway from reactants to products. To predict how well the reaction goes to products, we study its *thermodynamics*, the energetics of the reaction at equilibrium. The amounts of reactants and products present at equilibrium depend on their relative stabilities.

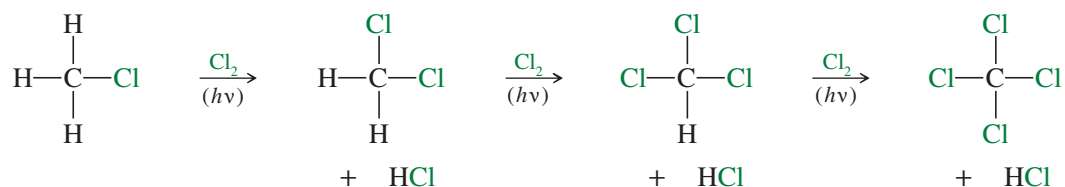
Even though the equilibrium may favor the formation of a product, the reaction may not take place at a useful rate. To use a reaction in a realistic time period (and to keep the reaction from becoming violent), we study its *kinetics*, the variation of reaction rates with different conditions and concentrations of reagents. Understanding the reaction's kinetics helps us to propose reaction mechanisms that are consistent with the behavior we observe.

4-2 Chlorination of Methane

The chlorination of methane is an important industrial reaction, with a relatively simple mechanism that illustrates many of the important principles of a reaction. The reaction of methane with chlorine produces a mixture of chlorinated products, whose composition depends on the amount of chlorine added and also on the reaction conditions. Either light or heat is needed for the reaction to take place at a useful rate. When chlorine is added to methane, the first reaction is



This reaction may continue; heat (Δ) or light ($h\nu$) is needed for each step:



This sequence raises several questions about the chlorination of methane. Why is heat or light needed for the reaction to go? Why do we get a mixture of products? Is there any way to modify the reaction to get just one pure product? Are the observed products formed because they are the most stable products possible? Or are they favored because they are formed faster than any other products?

The answers to these questions involve three aspects of the reaction: the mechanism, the thermodynamics, and the kinetics.

1. The **mechanism** is the complete, step-by-step description of exactly which bonds break and which bonds form, and in what order to give the observed products.
2. **Thermodynamics** is the study of the energy changes that accompany chemical and physical transformations. It allows us to compare the stability of reactants and products and predict which compounds are favored by the equilibrium.
3. **Kinetics** is the study of reaction rates, determining which products are formed fastest. Kinetics also helps to predict how the rate will change if we change the reaction conditions.

PROBLEM-SOLVING HINT

Free-radical reactions are some of the most common reactions in nature, but they are not the most common reactions used by organic chemists. The ionic reactions, covered in Chapters 6 and 7, are more common in the laboratory. We begin our study with free-radical reactions because they are not influenced by solvents or other variables that must be considered in the common ionic reactions.

We will use the chlorination of methane to show how to study a reaction. Before we can propose a detailed mechanism for the chlorination, we must learn everything we can about how the reaction works and what factors affect the reaction rate and the product distribution.

A careful study of the chlorination of methane has established three important characteristics:

1. *The chlorination does not occur at room temperature in the absence of light.* The reaction begins when light falls on the mixture or when it is heated. Thus, we know that this reaction requires some form of energy to *initiate* it.
2. *The most effective wavelength of light is a blue color that is strongly absorbed by chlorine gas.* This finding implies that absorbing light activates the chlorine molecule in some way so that it initiates the reaction with methane.
3. *The light-initiated reaction has a high quantum yield.* This means that many molecules of the product are formed for every photon of light absorbed. Our mechanism must explain how hundreds of individual reactions of methane with chlorine result from the absorption of a single photon by a single molecule of chlorine.

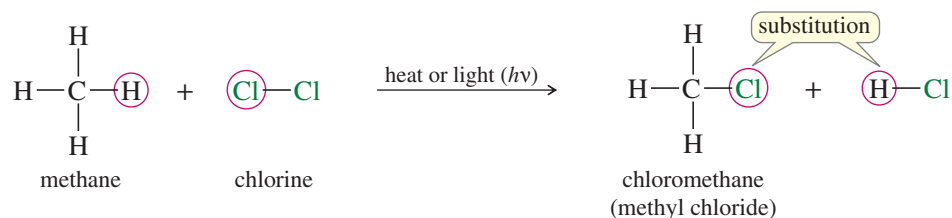
4-3 The Free-Radical Chain Reaction

A **chain reaction** mechanism has been proposed to explain the chlorination of methane.

A chain reaction consists of three kinds of steps:

1. The **initiation step**, which generates a reactive intermediate.
2. **Propagation steps**, in which the reactive intermediate reacts with a stable molecule to form a product and another reactive intermediate, allowing the chain to continue until the supply of reactants is exhausted or the reactive intermediate is destroyed.
3. **Termination steps**, side reactions that destroy reactive intermediates and tend to slow or stop the reaction.

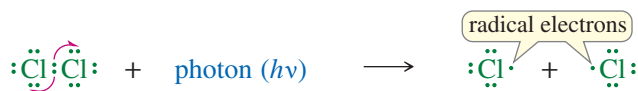
In studying the chlorination of methane, we will consider just the first reaction to form chloromethane (common name *methyl chloride*). This reaction is a **substitution**: Chlorine does not add to methane, but a chlorine atom *substitutes* for one of the hydrogen atoms, which becomes part of the HCl by-product.



4-3A The Initiation Step: Generation of Radicals

Blue light, absorbed by chlorine but not by methane, promotes this reaction. Therefore, initiation probably results from the absorption of light by a molecule of chlorine. Blue light has about the right energy to split a chlorine molecule (Cl_2) into two chlorine atoms, which requires 242 kJ/mol (58 kcal/mol).^{*} The splitting of a chlorine molecule by absorption of a photon is shown as follows:

Initiation: Formation of reactive intermediates.



^{*}The energy of a photon of light is related to its frequency ν by the relationship $E = h\nu$, where h is Planck's constant. Blue light has an energy of about 250 kJ (60 kcal) per einstein (an einstein is a mole of photons).

Application: Aging

Free radicals may play a role in diseases and accelerate aging. In the course of everyday life, reactive oxygen species are encountered in the environment and produced in the body. These compounds break down into short-lived hydroxyl radicals, which can react with the body's proteins and DNA. The resulting damage accumulates and may result in heart disease, cancer, and premature aging.

Notice the fishhook-shaped half-arrows used to show the movement of single unpaired electrons. Just as we use curved arrows to represent the movement of electron *pairs*, we use these curved half-arrows to represent the movement of single electrons. These half-arrows show that the two electrons in the Cl—Cl bond separate, and one leaves with each chlorine atom.

The splitting of a Cl₂ molecule is an initiation step that produces two highly reactive chlorine atoms. A chlorine atom is an example of a **reactive intermediate**, a short-lived species that is never present in high concentration because it reacts as quickly as it is formed. Each Cl· atom has an odd number of valence electrons (seven), one of which is unpaired. The unpaired electron is called the *odd electron* or the *radical electron*. Species with unpaired electrons are called **radicals** or **free radicals**. Radicals are electron-deficient because they lack an octet. The odd electron readily combines with an electron in another atom to complete an octet and form a bond. Figure 4-1 shows the Lewis structures of some free radicals. Radicals are often represented by a structure with a single dot representing the unpaired odd electron.

PROBLEM 4-1

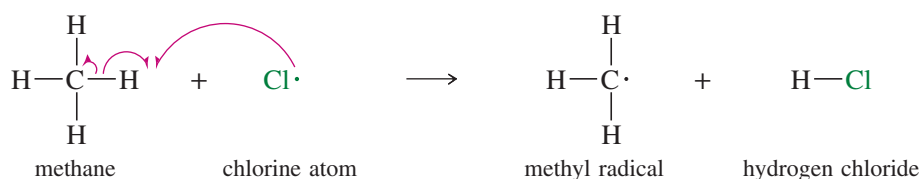
Draw Lewis structures for the following free radicals.

- (a) The ethyl radical, CH₃—CH₂· (b) The *tert*-butyl radical, (CH₃)₃C·
 (c) The isopropyl radical (2-propyl radical) (d) The iodine atom

Propagation: Formation of products with regeneration of reactive intermediates.

4-3B Propagation Steps

When a chlorine radical collides with a methane molecule, it abstracts (removes) a hydrogen atom from methane. One of the electrons in the C—H bond remains on carbon while the other combines with the odd electron on the chlorine atom to form the H—Cl bond.

First propagation step

This step forms only one of the final products: the molecule of HCl. A later step must form chloromethane. Notice that the first propagation step begins with one free radical (the chlorine atom) and produces another free radical (the methyl radical). The regeneration of a free radical is characteristic of a propagation step of a chain reaction. The reaction can continue because another reactive intermediate is produced.

FIGURE 4-1 Free radicals are reactive species with odd numbers of electrons. The unpaired electron is represented by a single unpaired dot in the formula.

Lewis structures*Written*

chlorine atom

bromine atom

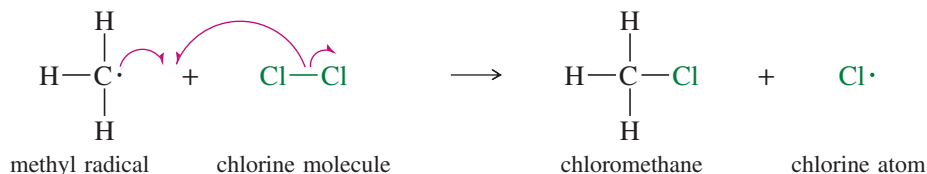
hydroxyl radical

methyl radical

ethyl radical

In the second propagation step, the methyl radical reacts with a molecule of chlorine to form chloromethane. The odd electron of the methyl radical combines with one of the two electrons in the Cl—Cl bond to give the Cl—CH₃ bond, and the chlorine atom is left with the odd electron.

Second propagation step



In addition to forming chloromethane, the second propagation step produces another chlorine radical. The chlorine radical can react with another molecule of methane, giving HCl and a methyl radical, which reacts with Cl₂ to give chloromethane and regenerate yet another chlorine radical. In this way, the chain reaction continues until the supply of the reactants is exhausted or some other reaction consumes the radical intermediates. The chain reaction explains why hundreds of molecules of methyl chloride and HCl are formed by each photon of light that is absorbed. We can summarize the reaction mechanism as follows.

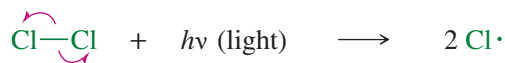


KEY MECHANISM 4-1 Free-Radical Halogenation

Like many other radical reactions, free-radical halogenation is a chain reaction. Chain reactions usually require one or more initiation steps to form radicals, followed by propagation steps that produce products and regenerate radicals.

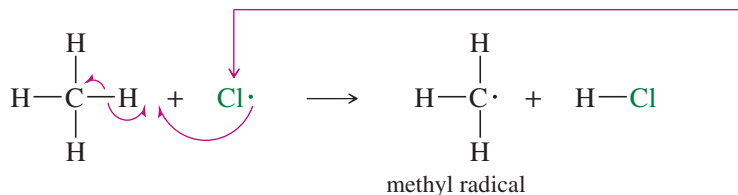
Initiation: Radicals are formed.

Light supplies the energy to split a chlorine molecule.

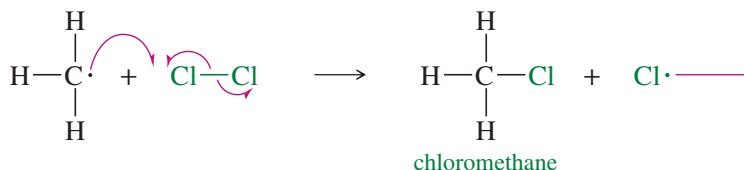


Propagation: A radical reacts to generate another radical.

Step 1: A chlorine radical abstracts a hydrogen to generate an alkyl radical.



Step 2: The alkyl radical reacts with Cl₂ to generate the product and a chlorine radical.

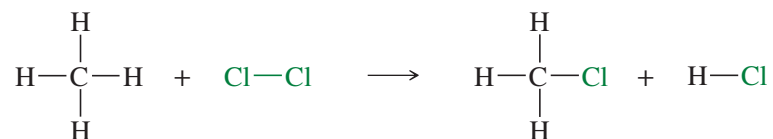


continues
the chain

The chlorine radical generated in step 2 goes on to react in another step 1, continuing the chain.

(continued)

The overall reaction is simply the sum of the propagation steps:



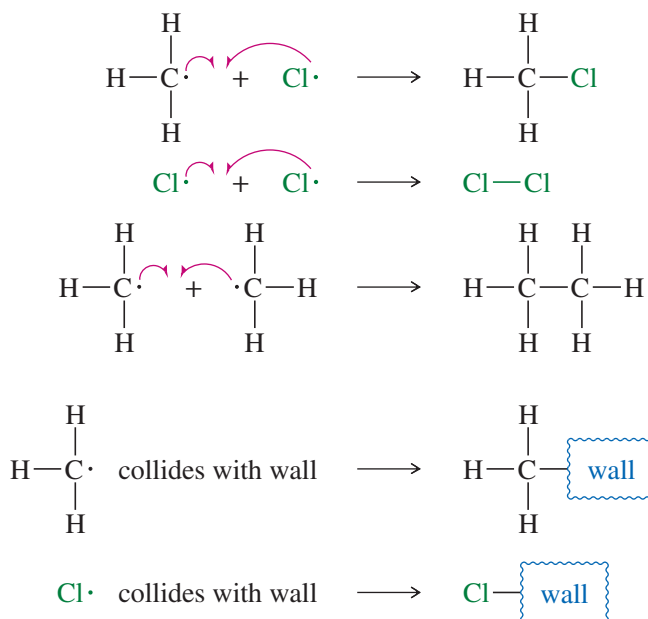
QUESTION: What factors characterize the propagation steps of a chain reaction?

PROBLEM 4-2

- Write the propagation steps leading to the formation of dichloromethane (CH_2Cl_2) from chloromethane.
- Explain why free-radical halogenation usually gives mixtures of products.
- How could an industrial plant control the proportions of methane and chlorine to favor production of CCl_4 ? To favor production of CH_3Cl ?

4-3C Termination Reactions

If anything happens to consume some of the free-radical intermediates without generating new ones, the chain reaction will slow or stop. Such a side reaction is called a **termination reaction**: a step that produces fewer reactive intermediates (free radicals) than it consumes. Following are some of the possible termination reactions in the chlorination of methane:



Application: Free-Radical Inhibitors

Some foods, additives, and supplements contain compounds that react with free radicals to terminate chain reactions. These radical inhibitors, called anti-oxidants, help to prevent the air-oxidation of foods by free-radical reactions. Current research is directed toward determining whether free-radical inhibitors in foods also can help cells to avoid damage caused by free radicals.

The combination of any two free radicals is a termination step because it decreases the number of free radicals. Other termination steps involve reactions of free radicals with the walls of the vessel or with any contaminants. Although the first of these termination steps gives chloromethane, one of the products, it consumes the free radicals that are necessary for the reaction to continue, thus breaking the chain. Its contribution to the amount of product obtained from the reaction is small compared with the contribution of the propagation steps.

While a chain reaction is in progress, the concentration of radicals is very low. The probability that two radicals will combine in a termination step is lower than the probability that each will encounter a molecule of reactant and give a propagation step. The termination steps become important toward the end of the reaction, when there are relatively few molecules of reactants available. At this point, the free radicals are less likely to encounter a molecule of reactant than they are to encounter each other (or the wall of the container). The chain reaction quickly stops.

PROBLEM 4-3

Each of the following proposed mechanisms for the free-radical chlorination of methane is wrong. Explain how the experimental evidence disproves each mechanism.

- (a) $\text{Cl}_2 + h\nu \rightarrow \text{Cl}_2^*$ (an “activated” form of Cl_2)
 $\text{Cl}_2^* + \text{CH}_4 \rightarrow \text{HCl} + \text{CH}_3\text{Cl}$
- (b) $\text{CH}_4 + h\nu \rightarrow \cdot\text{CH}_3 + \text{H}\cdot$
 $\cdot\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$
 $\text{Cl}\cdot + \text{H}\cdot \rightarrow \text{HCl}$

PROBLEM-SOLVING HINT

In a free-radical chain reaction, initiation steps generally create new free radicals. Propagation steps usually combine a free radical and a reactant to give a product and another free radical. Termination steps generally decrease the number of free radicals.

PROBLEM 4-4

Free-radical chlorination of hexane gives very poor yields of 1-chlorohexane, while cyclohexane can be converted to chlorocyclohexane in good yield.

- (a) How do you account for this difference?
- (b) What ratio of reactants (cyclohexane and chlorine) would you use for the synthesis of chlorocyclohexane?

4-4 Equilibrium Constants and Free Energy

Now that we have determined a mechanism for the chlorination of methane, we can consider the energetics of the individual steps. Let's begin by reviewing some of the principles needed for this discussion.

Thermodynamics is the branch of chemistry that deals with the energy changes accompanying chemical and physical transformations. These energy changes are most useful for describing the properties of systems at **equilibrium**. Let's review how energy and entropy variables describe an equilibrium.

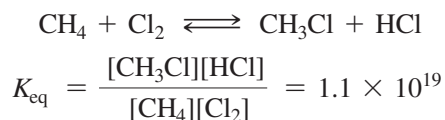
The equilibrium concentrations of reactants and products are governed by the **equilibrium constant** of the reaction. For example, if a moles of A and b moles of B react to give c moles of C and d moles of D, then the equilibrium constant K_{eq} is defined by the following equation:

$$a\text{A} + b\text{B} \rightleftharpoons c\text{C} + d\text{D}$$

$$K_{\text{eq}} = \frac{[\text{products}]}{[\text{reactants}]} = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$$

The value of K_{eq} tells us the position of the equilibrium: whether the products or the reactants are more stable, and therefore energetically favored. If K_{eq} is larger than 1, the reaction is favored as written from left to right. If K_{eq} is less than 1, the reverse reaction is favored (from right to left as written).

The chlorination of methane has a large equilibrium constant of about 1.1×10^{19} .



The equilibrium constant for chlorination is so large that the remaining amounts of the reactants are close to zero at equilibrium. Such a reaction is said to *go to completion*, and the value of K_{eq} is a measure of the reaction's tendency to *go to completion*.

From the value of K_{eq} , we can calculate the change in **free energy** (sometimes called **Gibbs free energy**) that accompanies the reaction. Free energy is represented by G , and the change (Δ) in free energy associated with a reaction is represented by ΔG , the difference between the free energy of the products and the free energy of the reactants. ΔG is a measure of the amount of energy available to do work.

$$\Delta G = (\text{free energy of products}) - (\text{free energy of reactants})$$

If the energy levels of the products are lower than the energy levels of the reactants (a “downhill” reaction), then the reaction is energetically favored; and this equation gives a negative value of ΔG , corresponding to a decrease in the energy of the system.

The **standard Gibbs free energy change**, ΔG° , is most commonly used. The symbol $^\circ$ designates a reaction involving reactants and products in their standard states (pure substances in their most stable states at 25 °C and 1 atm pressure). The relationship between ΔG° and K_{eq} is given by the expression

$$K_{\text{eq}} = e^{-\Delta G^\circ/RT}$$

or, conversely, by

$$\Delta G^\circ = -RT(\ln K_{\text{eq}}) = -2.303RT(\log_{10} K_{\text{eq}})$$

where

$R = 8.314 \text{ J/kelvin-mol}$ (1.987 cal/kelvin-mol), the gas constant

$T = \text{absolute temperature, in kelvins}^*$

$e = 2.718$, the base of natural logarithms

The value of RT at 25 °C is about 2.48 kJ/mol (0.592 kcal/mol).

The formula shows that a reaction is favored ($K_{\text{eq}} > 1$) if it has a *negative* value of ΔG° (energy is released). A reaction that has a positive value of ΔG° (energy must be added) is unfavorable. These predictions agree with our intuition that reactions should go from higher-energy states to lower-energy states, with a net decrease in free energy.

Some chemists use the terms *exergonic* and *endergonic* to describe changes in free energy.

Exergonic = having a negative ΔG° .

Endergonic = having a positive ΔG° .

PROBLEM-SOLVING HINT

A reaction with a negative ΔG is favorable.

A reaction with a positive ΔG is unfavorable.

Compounds with lower free energy are more stable than those with higher free energy.

SOLVED PROBLEM 4-1

Calculate the value of ΔG° for the chlorination of methane.

SOLUTION

We know the value of K_{eq} for this reaction, so we can use the equation relating the free energy change with the equilibrium constant.

$$\Delta G^\circ = -2.303RT(\log K_{\text{eq}})$$

$$K_{\text{eq}} \text{ for the chlorination is } 1.1 \times 10^{19}, \text{ and } \log K_{\text{eq}} = 19.04$$

At 25 °C (about 298 °K), the value of RT is

$$RT = (8.314 \text{ J/kelvin-mol})(298 \text{ kelvin}) = 2478 \text{ J/mol, or } 2.48 \text{ kJ/mol}$$

Substituting, we have

$$\Delta G^\circ = (-2.303)(2.478 \text{ kJ/mol})(19.04) = -108.7 \text{ kJ/mol } (-25.9 \text{ kcal/mol})$$

This is a large negative value for ΔG° , showing that this chlorination has a large driving force that pushes it toward completion.

*Absolute temperatures (in kelvins) are correctly given without a degree sign, as in the equation $25^\circ\text{C} = 298 \text{ K}$. We will include the degree sign, however, to distinguish absolute temperatures (K) from equilibrium constants (K) as in $25^\circ\text{C} = 298^\circ\text{K}$.

4-5A Enthalpy

The **change in enthalpy** (ΔH°) is the heat of reaction—the amount of heat evolved or consumed in the course of a reaction, usually given in kilojoules (or kilocalories) per mole. The enthalpy change is a measure of the relative strength of bonding in the products and reactants. Reactions tend to favor products with the lowest enthalpy (those with the strongest bonds).

PROBLEM-SOLVING HINT

Breaking a bond is always endothermic and absorbs energy.

Forming a bond is always exothermic and releases energy.

If weaker bonds are broken and stronger bonds are formed, heat is evolved and the reaction is **exothermic** (negative value of ΔH°). In an exothermic reaction, the enthalpy term makes a favorable negative contribution to ΔG° . If stronger bonds are broken and weaker bonds are formed, then energy is consumed in the reaction, and the reaction is **endothermic** (positive value of ΔH°). In an endothermic reaction, the enthalpy term makes an unfavorable positive contribution to ΔG° .

The value of ΔH° for the chlorination of methane is about -105.1 kJ/mol (-25.0 kcal/mol). This is a highly exothermic reaction, with the decrease in enthalpy serving as the primary driving force.

4-5B Entropy

Entropy is often described as a measure of randomness, disorder, or freedom of motion. Reactions tend to favor products with the greatest entropy. Notice the negative sign in the entropy term ($-T\Delta S^\circ$) of the free-energy expression. A positive value of the entropy change (ΔS°), indicating that the products have more freedom of motion than the reactants, makes a favorable (negative) contribution to ΔG° .

In many cases, the enthalpy change (ΔH°) is much larger than the entropy change (ΔS°), and the enthalpy term dominates the equation for ΔG° . Thus, a negative value of ΔS° does not necessarily mean that the reaction has an unfavorable value of ΔG° . The formation of strong bonds (the change in enthalpy) is usually the most important component in the driving force for a reaction.

In the chlorination of methane, the value of ΔS° is $+12.1$ J/kelvin-mole (2.89 cal/kelvin-mole). The $-T\Delta S^\circ$ term in the free energy is

$$\begin{aligned} -T\Delta S^\circ &= -(298^\circ\text{K})(12.1\text{ J/kelvin-mol}) = -3610\text{ J/mol} \\ &= -3.61\text{ kJ/mol } (-0.86\text{ kcal/mol}) \end{aligned}$$

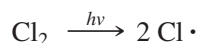
The value of $\Delta G^\circ = -108.7$ kJ/mol is divided into enthalpy and entropy terms:

$$\begin{aligned} \Delta G^\circ &= \Delta H^\circ - T\Delta S^\circ = -105.1\text{ kJ/mol} - 3.61\text{ kJ/mol} \\ &= -108.7\text{ kJ/mol } (-25.9\text{ kcal/mol}) \end{aligned}$$

The enthalpy change is the largest factor in the driving force for chlorination. This is the case in most organic reactions: The entropy term is often small in relation to the enthalpy term. When we discuss chemical reactions involving the breaking and forming of bonds, we can often use the values of the enthalpy changes (ΔH°), under the assumption that $\Delta G^\circ \cong \Delta H^\circ$. We must be cautious in making this approximation, however, because some reactions have relatively small changes in enthalpy and larger changes in entropy.

SOLVED PROBLEM 4-2

Predict whether the value of ΔS° for the dissociation of Cl_2 is positive (favorable) or negative (unfavorable). What effect does the entropy term have on the sign of the value of ΔG° for this reaction?

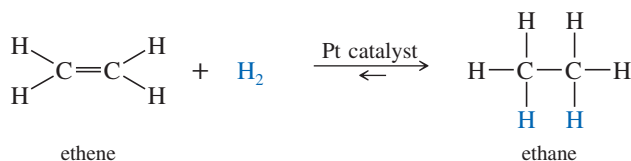


SOLUTION

Two isolated chlorine atoms have more freedom of motion than a single chlorine molecule. Therefore, the change in entropy is positive, and the entropy term ($-T\Delta S^\circ$) is negative. This negative (favorable) value of ($-T\Delta S^\circ$) is small, however, compared with the much larger, positive (unfavorable) value of ΔH° required to break the Cl—Cl bond. The chlorine molecule is much more stable than two chlorine atoms, showing that the positive enthalpy term predominates.

PROBLEM 4-7

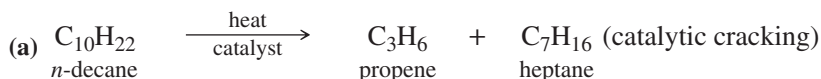
When ethene is mixed with hydrogen in the presence of a platinum catalyst, hydrogen adds across the double bond to form ethane. At room temperature, the reaction goes to completion. Predict the signs of ΔH° and ΔS° for this reaction. Explain these signs in terms of bonding and freedom of motion.

**PROBLEM-SOLVING HINT**

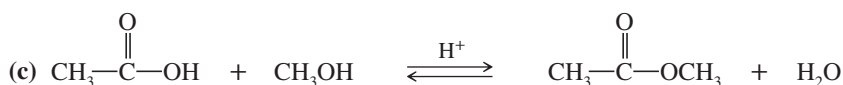
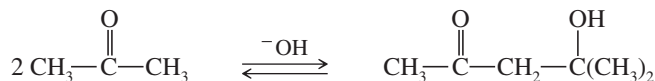
In general, two smaller molecules (or fragments, such as radicals) have more freedom of motion (greater entropy) than one larger molecule.

PROBLEM 4-8

For each reaction, estimate whether ΔS° for the reaction is positive, negative, or impossible to predict.



(b) The formation of diacetone alcohol:



4-6 Bond-Dissociation Enthalpies

We can put known amounts of methane and chlorine into a bomb calorimeter and use a hot wire to initiate the reaction. The temperature rise in the calorimeter is used to calculate the precise value of the heat of reaction, ΔH° . This measurement shows that 105 kJ (25 kcal) of heat is evolved (exothermic) for each mole of methane converted to chloromethane. Thus, ΔH° for the reaction is negative, and the heat of reaction is given as

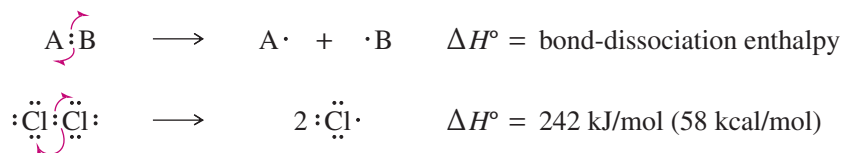
$$\Delta H^\circ = -105 \text{ kJ/mol } (-25 \text{ kcal/mol})$$

In many cases, we want to predict whether a particular reaction will be endothermic or exothermic, without actually measuring the heat of reaction. We can calculate an approximate heat of reaction by adding and subtracting the energies involved in the breaking and forming of bonds. To do this calculation, we need to know the energies of the affected bonds.

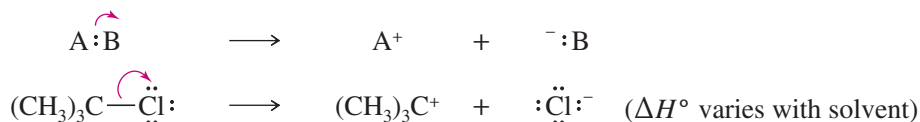
The **bond-dissociation enthalpy (BDE, also called bond-dissociation energy)** is the amount of enthalpy required to break a particular bond **homolytically**—that is, in such a way that each bonded atom retains one of the bond's two electrons. In contrast,

when a bond is broken **heterolytically** (as in an acid–base reaction), one of the atoms retains both electrons.

Homolytic cleavage (free radicals result)



Heterolytic cleavage (ions result)



Homolytic cleavage (radical cleavage) forms free radicals, while **heterolytic cleavage (ionic cleavage)** forms ions. Enthalpies for heterolytic (ionic) cleavage depend strongly on the solvent's ability to solvate the ions that result. Homolytic cleavage is used to define bond-dissociation enthalpies because the values do not vary so much with different solvents or with no solvent. Note that a curved arrow is used to show the movement of the electron pair in an ionic cleavage and that curved half-arrows are used to show the separation of individual electrons in a homolytic cleavage.

Energy is released when bonds are formed, and energy is consumed to break bonds. Therefore, bond-dissociation enthalpies are always positive (endothermic). The overall enthalpy change for a reaction is the sum of the dissociation enthalpies of the bonds broken minus the sum of the dissociation enthalpies of the bonds formed.

$$\Delta H^\circ = \Sigma(\text{BDE of bonds broken}) - \Sigma(\text{BDE of bonds formed})$$

For the hypothetical reaction



$$\Delta H^\circ = (\text{BDE of A}-\text{B}) + (\text{BDE of C}-\text{D}) - (\text{BDE of A}-\text{C}) - (\text{BDE of B}-\text{D})$$

By studying the heats of reaction for many different reactions, chemists have developed reliable tables of bond-dissociation enthalpies. Table 4-2 gives the bond-dissociation enthalpies for the homolysis of bonds in a variety of molecules.

4-7 Enthalpy Changes in Chlorination

We can use values from Table 4-2 to predict the heat of reaction for the chlorination of methane. This reaction involves the breaking (positive values) of a CH_3-H bond and a $\text{Cl}-\text{Cl}$ bond, and the formation (negative values) of a CH_3-Cl bond and a $\text{H}-\text{Cl}$ bond.

Overall reaction



<i>Bonds broken ΔH° (per mole)</i>		<i>Bonds formed ΔH° (per mole)</i>	
$\text{Cl}-\text{Cl}$	+240 kJ (+57 kcal)	$\text{H}-\text{Cl}$	−432 kJ (−103 kcal)
CH_3-H	+439 kJ (+105 kcal)	CH_3-Cl	−350 kJ (−84 kcal)
Total	+679 kJ (+162 kcal)	Total	−782 kJ (−187 kcal)

$$\Delta H^\circ = +679 \text{ kJ/mol} + (-782) \text{ kJ/mol} = -103 \text{ kJ/mol} (-25 \text{ kcal/mol})$$

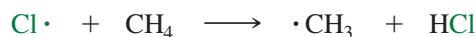
TABLE 4-2 Bond-Dissociation Enthalpies for Homolytic Cleavages

A:B $\longrightarrow$ A· + ·B					
Bond	Bond-Dissociation Enthalpy		Bond	Bond-Dissociation Enthalpy	
	kJ/mol	kcal/mol		kJ/mol	kcal/mol
H—X bonds and X—X bonds			Bonds to secondary carbons		
H—H	436	104	(CH ₃) ₂ CH—H	413	99
D—D	440	105	(CH ₃) ₂ CH—F	463	111
F—F	154	37	(CH ₃) ₂ CH—Cl	356	85
Cl—Cl	240	57	(CH ₃) ₂ CH—Br	309	74
Br—Br	190	45	(CH ₃) ₂ CH—I	238	57
I—I	149	36	(CH ₃) ₂ CH—OH	399	95
H—F	570	136	Bonds to tertiary carbons		
H—Cl	432	103	(CH ₃) ₃ C—H	403	96
H—Br	366	88	(CH ₃) ₃ C—F	464*	111*
H—I	298	71	(CH ₃) ₃ C—Cl	355	85
HO—H	497	119	(CH ₃) ₃ C—Br	304	73
HS—H	382	91	(CH ₃) ₃ C—I	233	56
HOO—H	367	88	(CH ₃) ₃ C—OH	401	96
HO—OH	213	51	Other C—H bonds		
H ₂ N—H	450	108	PhCH ₂ —H (benzylic)	376	90
Methyl bonds			H ₂ C=CHCH ₂ —H (allylic)	372	89
CH ₃ —H	439	105	H ₂ C=CH—H (vinyl)	463	111
CH ₃ —F	481	115	Ph—H (aromatic)	472	113
CH ₃ —Cl	350	84	HC≡C—H (acetylenic)	558	133
CH ₃ —Br	302	72	C—C bonds		
CH ₃ —I	241	58	CH ₃ —CH ₃	377	90
CH ₃ —OH	385	92	CH ₃ CH ₂ —CH ₃	372	89
CH ₃ —NH ₂	356	85	CH ₃ CH ₂ —CH ₂ CH ₃	368	88
Bonds to primary carbons			(CH ₃) ₂ CH—CH ₃	371	89
CH ₃ CH ₂ —H	423	101	(CH ₃) ₃ C—CH ₃	366	87
CH ₃ CH ₂ —F	464*	111*	C=C and C=X bonds		
CH ₃ CH ₂ —Cl	355	85	H ₂ C=CH ₂	728	174
CH ₃ CH ₂ —Br	303	72	H ₂ C=NH	736*	176
CH ₃ CH ₂ —I	238	57	H ₂ C=O	749	179
CH ₃ CH ₂ —OH	393	94			
CH ₃ CH ₂ CH ₂ —H	423	101			
CH ₃ CH ₂ CH ₂ —F	464*	111*			
CH ₃ CH ₂ CH ₂ —Cl	355	85			
CH ₃ CH ₂ CH ₂ —Br	303	72			
CH ₃ CH ₂ CH ₂ —I	238	57			

*approximate

The bond-dissociation enthalpies also provide the heat of reaction for each individual step:

First propagation step



Breaking a $\text{CH}_3\text{—H}$ bond	+439 kJ/mol (+105 kcal/mol)
Forming a H—Cl bond	−432 kJ/mol (−103 kcal/mol)
Step total	+7 kJ/mol (+2 kcal/mol)

Second propagation step



Breaking a Cl—Cl bond	+240 kJ/mol (+57 kcal/mol)
Forming a $\text{CH}_3\text{—Cl}$ bond	−350 kJ/mol (−84 kcal/mol)
Step total	−110 kJ/mol (−27 kcal/mol)

$$\text{Grand total} = +7 \text{ kJ/mol} + (-110 \text{ kJ/mol}) = -103 \text{ kJ/mol} (-25 \text{ kcal/mol})$$

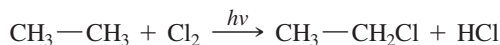
PROBLEM-SOLVING HINT

Bond-dissociation enthalpies are for breaking bonds, which costs energy. In calculating values of ΔH° , use positive BDE values for bonds that are broken and negative values for bonds that are formed.

The sum of the values of ΔH° for the individual propagation steps gives the overall enthalpy change for the reaction. The initiation step, $\text{Cl}_2 \rightarrow 2 \text{Cl}\cdot$, is not added to give the overall enthalpy change because it is not necessary for each molecule of product formed. The first splitting of a chlorine molecule simply begins the chain reaction, which generates hundreds or thousands of molecules of chloromethane. The energy needed to break the Cl—Cl bond is already included in the second propagation step.

PROBLEM 4-9

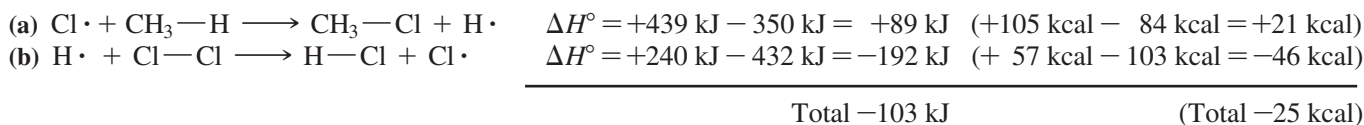
(a) Propose a mechanism for the free-radical chlorination of ethane,



(b) Calculate ΔH° for each step in this reaction.

(c) Calculate the overall value of ΔH° for this reaction.

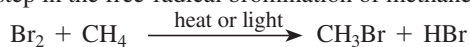
Alternative Mechanism The mechanism we have used is not the only one that might be proposed to explain the reaction of methane with chlorine. We know that the initiating step must be the splitting of a molecule of Cl_2 , but there are other propagation steps that would form the correct products:



This alternative mechanism seems plausible, but step (a) is endothermic by 89 kJ/mol (21 kcal/mol). Reactions generally follow the lowest-energy pathway available, and the previous mechanism provides a lower-energy alternative. When a chlorine atom collides with a methane molecule, it will not react to give methyl chloride and a hydrogen atom ($\Delta H^\circ = +89 \text{ kJ} = +21 \text{ kcal}$); it will react to give HCl and a methyl radical ($\Delta H^\circ = +7 \text{ kJ} = +2 \text{ kcal}$), the first propagation step of the correct mechanism.

PROBLEM 4-10

(a) Using bond-dissociation enthalpies from Table 4-2 (page 209), calculate the heat of reaction for each step in the free-radical bromination of methane.



(b) Calculate the overall heat of reaction.

4-8 Kinetics and the Rate Equation

Kinetics is the study of reaction rates. How fast a reaction goes is just as important as the position of its equilibrium. Just because thermodynamics favors a reaction (negative ΔG°) does not necessarily mean the reaction will actually occur. For example, a mixture of gasoline and oxygen does not react without a spark or a catalyst. Similarly, a mixture of methane and chlorine does not react if it is kept cold and dark.

The **rate of a reaction** is a measure of how fast the products appear and the reactants disappear. We can determine the rate by measuring the increase in the concentrations of the products with time, or the decrease in the concentrations of the reactants with time.

Reaction rates depend on the concentrations of the reactants. The greater the concentrations, the more often the reactants collide and the greater the chance of reaction. A **rate equation** (sometimes called a **rate law**) is the relationship between the concentrations of the reactants and the observed reaction rate. Each reaction has its own rate equation, *determined experimentally* by changing the concentrations of the reactants and measuring the change in the rate. For example, consider the general reaction



The reaction rate is usually proportional to the concentrations of the reactants ($[A]$ and $[B]$) raised to some powers, a and b . We can use a general rate expression to represent this relationship as

$$\text{rate} = k_r[A]^a[B]^b$$

where k_r is the **rate constant**, and the values of the powers (a and b) must be determined experimentally. We cannot guess or calculate the rate equation from just the stoichiometry of the reaction. The rate equation depends on the mechanism of the reaction and on the rates of the individual steps.

In the general rate equation, the power a is called the **order** of the reaction with respect to reactant A, and b is the order of the reaction with respect to B. The sum of these powers, $(a + b)$, is called the **overall order** of the reaction.

The following reaction has a simple rate equation:

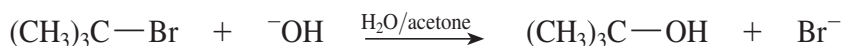


Experiments show that doubling the concentration of methyl bromide, $[\text{CH}_3\text{Br}]$, doubles the rate of reaction. Doubling the concentration of hydroxide ion, $[^-\text{OH}]$, also doubles the rate. Thus, the rate is proportional to both $[\text{CH}_3\text{Br}]$ and $[^-\text{OH}]$, so the rate equation has the following form:

$$\text{rate} = k_r[\text{CH}_3\text{Br}][^-\text{OH}]$$

This rate equation is *first order* in each of the two reagents because it is proportional to the first power of their concentrations. The rate equation is *second order overall* because the sum of the powers of the concentrations in the rate equation is 2; that is, (first order) + (first order) = second order overall.

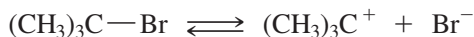
Reactions of the same overall type do not necessarily have the same form of rate equation. For example, the following similar reaction has a different kinetic order:



Doubling the concentration of *tert*-butyl bromide $[(\text{CH}_3)_3\text{C—Br}]$ causes the rate to double, but doubling the concentration of hydroxide ion $[^-\text{OH}]$ has no effect on the rate of this particular reaction. The rate equation is

$$\text{rate} = k_r[(\text{CH}_3)_3\text{C—Br}]$$

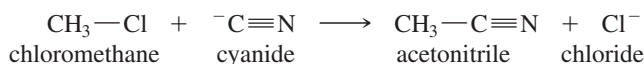
This reaction is first order in *tert*-butyl bromide, and zeroth order in hydroxide ion (proportional to $[\text{OH}^-]$ to the zeroth power). It is first order overall. This reaction is zeroth order in hydroxide ion because the slow step involves only *tert*-butyl bromide and not hydroxide ion:



The most important fact to remember is that *the rate equation must be determined experimentally*. We cannot predict the form of the rate equation from the stoichiometry of the reaction. We determine the rate equation experimentally, then use that information to propose consistent mechanisms.

SOLVED PROBLEM 4-3

Chloromethane reacts with dilute sodium cyanide ($\text{Na}^+ \text{CN}^-$) according to the following equation:



When the concentration of chloromethane is doubled, the rate is observed to double. When the concentration of cyanide ion is tripled, the rate is observed to triple.

- What is the kinetic order with respect to chloromethane?
- What is the kinetic order with respect to cyanide ion?
- What is the kinetic order overall?
- Write the rate equation for this reaction.

SOLUTION

- When $[\text{CH}_3\text{Cl}]$ is doubled, the rate doubles, which is 2 to the first power. The reaction is first order with respect to chloromethane.
- When $[\text{CN}^-]$ is tripled, the reaction rate triples, which is 3 to the first power. The reaction is first order with respect to cyanide ion.
- First order plus first order equals second order overall.
- $\text{rate} = k_r[\text{CH}_3\text{Cl}][\text{CN}^-]$

PROBLEM 4-11

The reaction of *tert*-butyl chloride with methanol



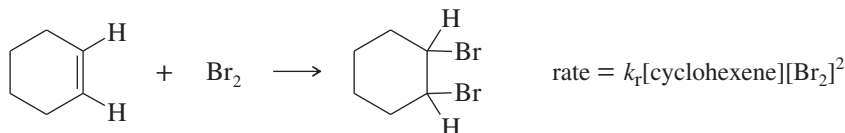
is found to follow the rate equation

$$\text{rate} = k_r[(\text{CH}_3)_3\text{C}-\text{Cl}]$$

- What is the kinetic order with respect to *tert*-butyl chloride?
- What is the kinetic order with respect to methanol?
- What is the kinetic order overall?

PROBLEM 4-12

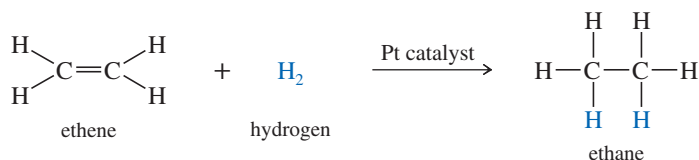
Under certain conditions, the bromination of cyclohexene follows an unusual rate law:



- What is the kinetic order with respect to cyclohexene?
- What is the kinetic order with respect to bromine?
- What is the overall kinetic order?

PROBLEM 4-13

When a small piece of platinum is added to a mixture of ethene and hydrogen, the following reaction occurs:



Doubling the concentration of hydrogen has no effect on the reaction rate. Doubling the concentration of ethene also has no effect.

- What is the kinetic order of this reaction with respect to ethene? With respect to hydrogen? What is the overall order?
- Write the unusual rate equation for this reaction.
- Explain this strange rate equation, and suggest what one might do to accelerate the reaction.

4-9 Activation Energy and the Temperature Dependence of Rates

Each reaction has its own characteristic rate constant, k_r . Its value depends on the conditions of the reaction, especially the temperature. This temperature dependence is expressed by the *Arrhenius equation*,

$$k_r = Ae^{-E_a/RT}$$

where

A = a constant (the “frequency factor”)

E_a = activation energy

R = the gas constant, 8.314 J/kelvin-mole (1.987 cal/kelvin-mole)

T = the absolute temperature

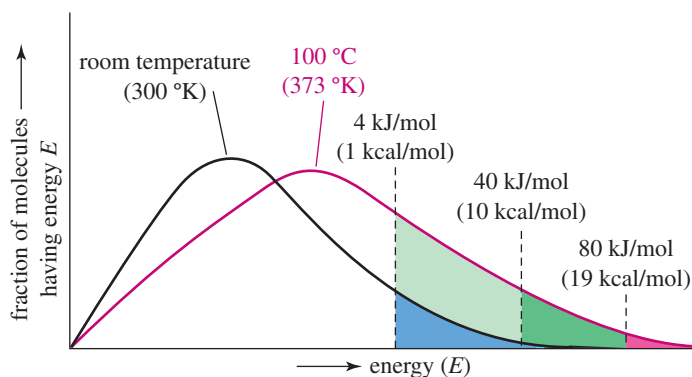
The **activation energy**, E_a , is the minimum kinetic energy the molecules must have to overcome the repulsions between their electron clouds when they collide. The exponential term $e^{-E_a/RT}$ corresponds to the fraction of collisions in which the particles have the minimum energy E_a needed to react. We can calculate E_a for a reaction by measuring how k_r varies with temperature, and substituting into the Arrhenius equation.

The *frequency factor* A accounts for the frequency of collisions and the fraction of collisions with the proper orientation for the reaction to occur. In most cases, only a small fraction of collisions occur between molecules with enough speed and with just the right orientation for reaction to occur. Far more collisions occur without enough kinetic energy or without the proper orientation, and the molecules simply bounce off each other.

The Arrhenius equation implies that the rate of a reaction depends on the fraction of collisions with kinetic energy of at least E_a . Figure 4-2 shows how the distribution of kinetic energies in a sample of a gas depends on the temperature. The black curved line shows the molecular energy distribution at room temperature, and the dashed lines show the energy needed to overcome barriers of 4 kJ/mol (1 kcal/mol), 40 kJ/mol (10 kcal/mol), and 80 kJ/mol (19 kcal/mol). The area under the curve to the right of each barrier corresponds to the fraction of molecules with enough energy to overcome that barrier.

The red curve shows how the energy distribution is shifted at 100 °C. At 100 °C, many more molecules have the energy needed to overcome the energy barriers, especially the 80 kJ/mol barrier. For smaller temperature changes, chemists often use an approximation: for reactions with typical activation energies of about 40 to 60 kJ/mol (10 to 15 kcal/mol), the reaction rate approximately doubles when the temperature is raised by 10 °C, as from 27 °C (near room temperature) to 37 °C (body temperature).

FIGURE 4-2 The dependence of kinetic energies on temperature. This graph shows how the fraction of molecules with a given activation energy decreases as the activation energy increases. At a higher temperature (red curve), more collisions have the needed energy.



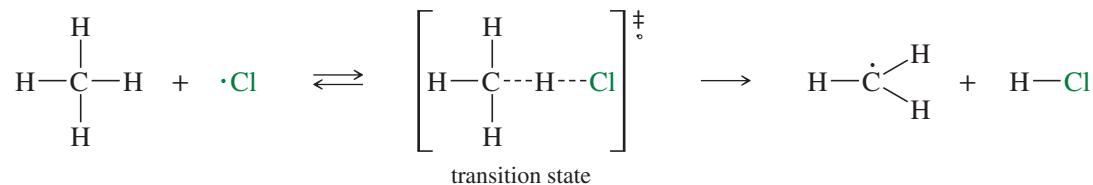
Because the rate constant increases quickly when the temperature is raised, it might seem that raising the temperature would always be a good way to save time by making reactions go faster. The problem with raising the temperature is that *all* reactions are accelerated, including all the unwanted side reactions. We try to find a temperature that allows the desired reaction to go at a reasonable rate without producing unacceptable rates of side reactions.

4-10 Transition States

The activation energy E_a represents the energy difference between the reactants and the **transition state**, the highest-energy state in a molecular collision that leads to reaction. In effect, the activation energy is the barrier that must be overcome for the reaction to take place. The value of E_a is always positive, and its magnitude depends on the relative energy of the transition state. The term *transition state* implies that this configuration is the transition between the reactants and products, and the molecules can either go on to products or return to reactants.

Unlike the reactants or products, a transition state is unstable and cannot be isolated. It is not an intermediate, because an **intermediate** is a species that exists for some finite length of time, even if it is very short. An intermediate has at least some stability, but the transition state is a transient on the path from one intermediate to another. The transition state is often symbolized by a superscript double dagger ($\ddagger$), and the changes in variables such as free energy, enthalpy, and entropy involved in achieving the transition state are symbolized $\Delta G^\ddagger$, $\Delta H^\ddagger$, and $\Delta S^\ddagger$. $\Delta G^\ddagger$ is similar to E_a , and the symbol $\Delta G^\ddagger$ is often used in speaking of the activation energy.

Transition states have high energies because bonds must begin to break before other bonds can form. The following equation shows the reaction of a chlorine radical with methane. The transition state shows the C—H bond partially broken and the H—Cl bond partially formed. Transition states are often enclosed by brackets to emphasize their transient nature. The double dagger next to the bracket means that this is a transition state, and the dot means that it contains a radical (unpaired) electron.



Reaction-Energy Diagrams The concepts of transition state and activation energy are easier to understand graphically. Figure 4-3 shows a **reaction-energy diagram** for a one-step exothermic reaction. The vertical axis of the energy diagram represents the total potential energy of all the species involved in the reaction. The horizontal axis is called the **reaction coordinate**. The reaction coordinate symbolizes the progress of the reaction, going from the reactants on the left to the products on the right.

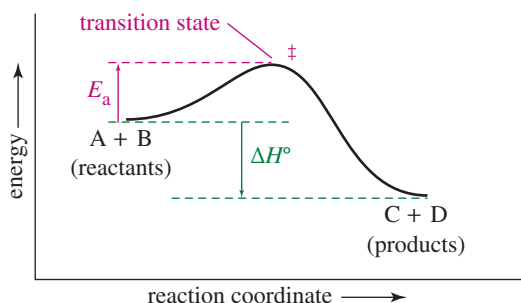


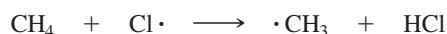
FIGURE 4-3 Reaction-energy diagram for a one-step exothermic reaction. The reactants are toward the left, and the products are toward the right. The vertical axis represents the potential energy. The transition state is the highest point on the graph, and the activation energy is the energy difference between the reactants and the transition state.

The transition state is the highest point on the graph, and the activation energy is the energy difference between the reactants and the transition state. The heat of reaction (ΔH°) is the difference in energy between the reactants and the products.

If a **catalyst** were added to the reaction in Figure 4-3, it would create a transition state of lower energy, thereby lowering the activation energy and increasing the reaction rate. Addition of a catalyst would not change the energies of the reactants and products, however, so the heat of reaction and the equilibrium constant would be unaffected.

SOLVED PROBLEM 4-4

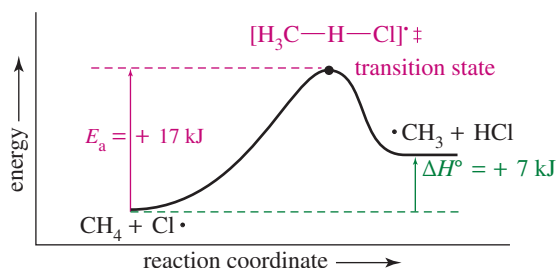
Consider the following reaction:



This reaction has an activation energy (E_a) of +17 kJ/mol (+4 kcal/mol) and a ΔH° of +7 kJ/mol (+2 kcal/mol). Draw a reaction-energy diagram for this reaction.

SOLUTION

We draw a diagram that shows the products to be 7 kJ *higher* in energy than the reactants. The barrier is made to be 17 kJ higher in energy than the reactants.



PROBLEM 4-14

(a) Draw the reaction-energy diagram for the following reverse reaction:



(b) What is the activation energy for this reverse reaction?

(c) What is the heat of reaction (ΔH°) for this reverse reaction?

PROBLEM 4-15

(a) Draw a reaction-energy diagram for the following reaction:



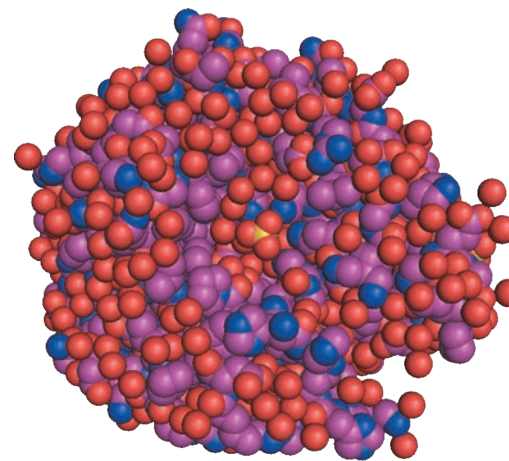
The activation energy is 4 kJ/mol (1 kcal/mol), and the overall ΔH° for the reaction is −110 kJ/mol (−27 kcal/mol).

(b) Give the equation for the reverse reaction.

(c) What is the activation energy for the reverse reaction?

Application: Biochemistry

Enzymes serve as biological catalysts. They speed up reactions without changing the energies of the reactants (called substrates) and products. Without enzymes, most of the reactions in our cells would not go fast enough to keep us alive. The enzyme shown is carbonic anhydrase, which increases the rate of the reaction $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$ by a factor of 7.7×10^6 . This rate enhancement corresponds to a reduction in E_a of 40 kJ/mol (9.5 kcal/mol).



4-11 Rates of Multistep Reactions

Many reactions proceed by mechanisms involving several steps and several intermediates. As we saw in Section 4-7, for example, the reaction of methane with chlorine goes through two propagation steps. The propagation steps are shown here, along with their heats of reaction and their activation energies. Just the propagation steps are shown because the rate of the initiation step is controlled by the amount of light or heat available to split chlorine molecules.

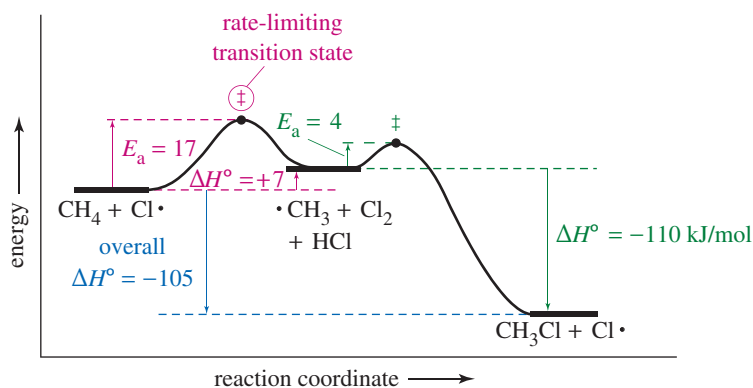
Step	ΔH° (per mole)	E_a (per mole)
$\text{CH}_4 + \text{Cl}\cdot \longrightarrow \cdot\text{CH}_3 + \text{HCl}$	+7 kJ (+2 kcal)	17 kJ (4 kcal)
$\cdot\text{CH}_3 + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}\cdot$	-110 kJ (-27 kcal)	4 kJ (1 kcal)

In this reaction, $\text{Cl}\cdot$ and $\text{CH}_3\cdot$ are *reactive intermediates*. Unlike transition states, reactive intermediates are stable as long as they do not collide with other atoms or molecules. As free radicals, however, $\text{Cl}\cdot$ and $\cdot\text{CH}_3$ are quite reactive toward other molecules. Figure 4-4 shows a single reaction-energy profile that includes both propagation steps of the chlorination. The energy maxima (high points) are the unstable transition states, and the energy minima (low points) are the intermediates. This complete energy profile provides most of the important information about the energetics of the reaction.

The Rate-Limiting Step In a multistep reaction, each step has its own characteristic rate. There can be only one overall reaction rate, however, and it is controlled by the **rate-limiting step** (also called the **rate-determining step**). In general, the *highest-energy* step of a multistep reaction is the “bottleneck,” and it determines the overall rate. How can we tell which step is rate limiting? If we have the reaction-energy diagram, it is simple: The highest point in the energy diagram is the transition state with the highest energy—generally the transition state for the rate-limiting step.

The highest point in the energy diagram of the chlorination of methane (Figure 4-4) is the transition state for the reaction of methane with a chlorine radical. This step must be rate limiting. If we calculate a rate for this slow step, it will be the rate for the overall reaction. The second, faster step will consume the products of the slow step as fast as they are formed.

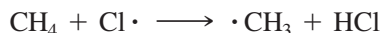
FIGURE 4-4 Combined reaction-energy diagram for the chlorination of methane. (Units are kJ/mol.) The energy maxima are transition states, and the energy minima are intermediates. $\text{Cl}\cdot$ and $\cdot\text{CH}_3$ are the reactive intermediates in this mechanism.



4-12 Temperature Dependence of Halogenation

We now apply what we know about rates to the reaction of methane with halogens. The rate-limiting step for chlorination is the endothermic reaction of the chlorine atom with methane to form a methyl radical and a molecule of HCl.

Rate-limiting step



The activation energy for this step is 17 kJ/mol (4 kcal/mol). At room temperature, the value of $e^{-E_a/RT}$ is 1300×10^{-6} . This value represents a rate that is fast but controllable.

In a free-radical chain reaction, every propagation step must occur quickly; otherwise, the free radicals will undergo unproductive collisions and participate in termination steps. We can predict how quickly the various halogen atoms react with methane given relative rates based on the measured activation energies of the slowest steps:

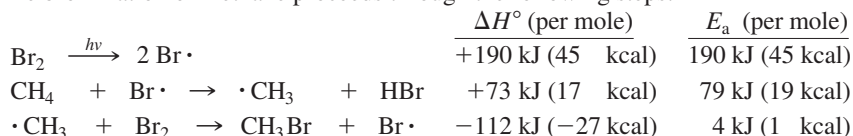
Reaction	E_a (per mole)	Relative Rate ($e^{-E_a/RT} \times 10^6$)	
		27 °C (300 °K)	227 °C (500 °K)
$\text{F}\cdot + \text{CH}_4 \longrightarrow \text{HF} + \cdot\text{CH}_3$	5 kJ (1.2 kcal)	140,000	300,000
$\text{Cl}\cdot + \text{CH}_4 \longrightarrow \text{HCl} + \cdot\text{CH}_3$	17 kJ (4 kcal)	1300	18,000
$\text{Br}\cdot + \text{CH}_4 \longrightarrow \text{HBr} + \cdot\text{CH}_3$	75 kJ (18 kcal)	9×10^{-8}	0.015
$\text{I}\cdot + \text{CH}_4 \longrightarrow \text{HI} + \cdot\text{CH}_3$	140 kJ (34 kcal)	2×10^{-19}	2×10^{-9}

These relative rates suggest how easily and quickly methane reacts with the different halogen radicals. The reaction with fluorine should be difficult to control because its rate is very high. Chlorine should react moderately at room temperature, but it may become difficult to control if the temperature rises much (the rate at 500 °K is rather high). The reaction with bromine is very slow, but heating might give an observable rate. Iodination is probably out of the question because its rate is exceedingly slow, even at 500 °K.

Laboratory halogenations show that our predictions are right. In fact, fluorine reacts explosively with methane, and chlorine reacts at a moderate rate. A mixture of bromine and methane must be heated to react, and iodine does not react at all.

PROBLEM 4-16

The bromination of methane proceeds through the following steps:



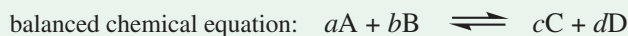
- Draw a complete reaction-energy diagram for this reaction.
- Label the rate-limiting step.
- Draw the structure of each transition state.
- Compute the overall value of ΔH° for the bromination.

PROBLEM 4-17

- Using the BDEs in Table 4-2 (page 209), compute the value of ΔH° for each step in the iodination of methane.
- Compute the overall value of ΔH° for iodination.
- Suggest *two* reasons why iodine does not react well with methane.

FOCUS Reaction Kinetics and Thermodynamics

The energetics of chemical reactions can be divided into **kinetics** and **thermodynamics**. They are determined in different ways, and tell us different things about reactions.



A and B are reactants; C and D are products; a , b , c , and d are coefficients

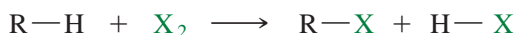
	KINETICS	THERMODYNAMICS
Gives what information?	how fast a reaction occurs; does not need to be an equilibrium	whether products or reactants are favored at equilibrium
Determined by?	activation energy, E_a , the highest energy barrier between reactants and products	the energy difference between reactants and products, ΔG
Does the reaction path matter?	YES! Kinetics is a <i>path function</i> .	NO! Thermodynamics is a <i>state function</i> ; only the starting and ending points matter.
Measured how?	Activation energy and rate laws can be determined only by experiment.	Equilibrium constant expression can be determined from the balanced equation. Bond energy differences can be calculated from tables to give ΔH values that approximate ΔG .
Math	rate law: $\text{reaction rate} = K_r [A]^x [B]^y$ where K_r is the rate constant, $[A]$ and $[B]$ are the concentrations of reactants, and x and y are the order of the rate expression in each reactant. The order of each reactant is helpful in determining the role of each reactant in the rate-limiting step of the mechanism.	$K_{\text{eq}} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$ from balanced chemical equation Equilibrium constant expression can be determined from the balanced equation. Bond energy differences can be calculated from tables.
Does temperature matter?	YES! Virtually all reactions go faster at higher temperature.	YES! Equilibrium depends on temperature, but not always in an obvious way.

4-13 Selectivity in Halogenation

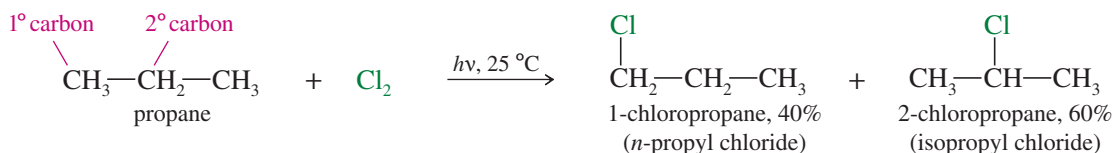
Up to now, we have limited our discussions to the halogenation of methane. Beginning our study with such a simple compound allowed us to concentrate on the thermodynamics and kinetics of the reaction. Now we consider halogenation of the “higher” alkanes, meaning those of higher molecular weight.

4-13A Chlorination of Propane: Product Ratios

Halogenation is a substitution, where a halogen atom replaces a hydrogen.



In methane, all four hydrogen atoms are identical, and it does not matter which hydrogen is replaced. In the higher alkanes, replacement of different hydrogen atoms may lead to different products. In the chlorination of propane, for example, two monochlorinated (just one chlorine atom) products are possible. One has the chlorine atom on a primary carbon atom, and the other has the chlorine atom on the secondary carbon atom.



The product ratio shows that replacement of hydrogen atoms by chlorine is not random. Propane has six primary hydrogens (hydrogens bonded to primary carbons) and only two secondary hydrogens (bonded to the secondary carbon), yet the major product results from substitution of a secondary hydrogen. We can calculate how reactive each kind of hydrogen is by dividing the amount of product observed by the number of hydrogens that can be replaced to give that product.

Figure 4-5 shows the definition of primary, secondary, and tertiary hydrogens and the calculation of their relative reactivity. Replacing either of the two secondary hydrogens accounts for 60% of the product, and replacing any of the six primary hydrogens accounts for 40% of the product. We calculate that each secondary hydrogen is 4.5 times as reactive as each primary hydrogen.

To explain this preference for reaction at the secondary position, we must look carefully at the reaction mechanism, as shown in Figure 4-6. When a chlorine atom reacts with propane, abstraction of a hydrogen atom can give either a primary radical

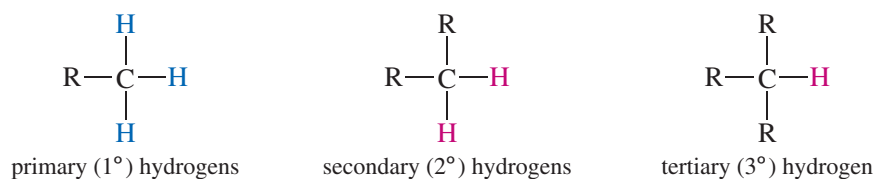
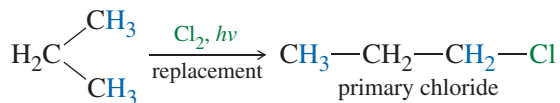


FIGURE 4-5 Definitions of primary, secondary, and tertiary hydrogens. There are six primary hydrogens in propane and only two secondary hydrogens, yet the major product results from replacement of a secondary hydrogen.

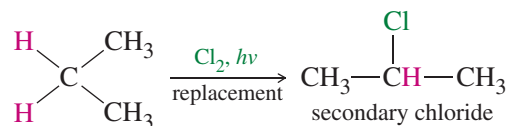
Six primary (1°) hydrogens



relative reactivity

$$\frac{40\%}{6 \text{ hydrogens}} = 6.67\% \text{ per H}$$

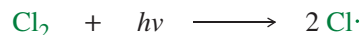
Two secondary (2°) hydrogens



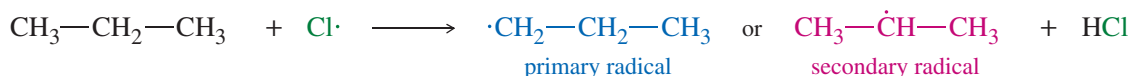
$$\frac{60\%}{2 \text{ hydrogens}} = 30.0\% \text{ per H}$$

The 2° hydrogens are $\frac{30.0}{6.67} = 4.5$ times as reactive as the 1° hydrogens.

Initiation: Splitting of the chlorine molecule



First propagation step: Abstraction (removal) of a primary or secondary hydrogen



Second propagation step: Reaction with chlorine to form the alkyl chloride

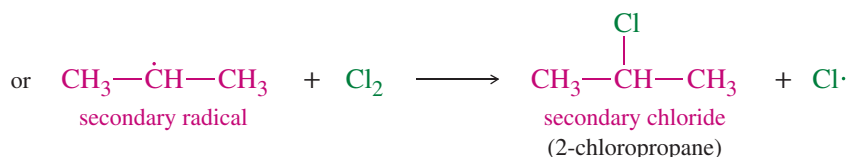
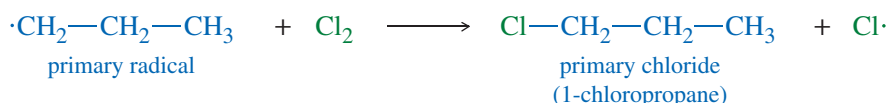


FIGURE 4-6 The mechanism for free-radical chlorination of propane. The first propagation step forms either a primary radical or a secondary radical. This radical determines whether the final product will be the primary alkyl chloride or the secondary alkyl chloride.

or a secondary radical. The structure of the radical formed in this step determines the structure of the observed product, either 1-chloropropane or 2-chloropropane. The product ratio shows that the secondary radical is formed preferentially. This preference for reaction at the secondary position results from the greater stability of the secondary free radical and the transition state leading to it. We explain this preference in more detail in the next section.

PROBLEM 4-18

What would be the product ratio in the chlorination of propane if all the hydrogens were abstracted at equal rates?

PROBLEM 4-19

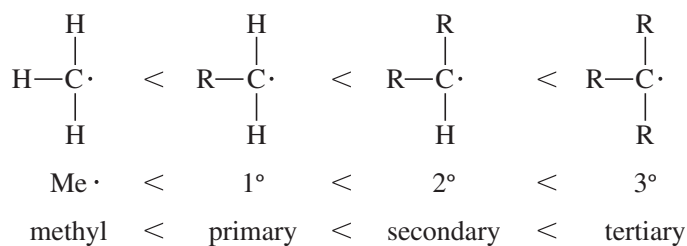
Classify each hydrogen atom in the following compounds as primary (1°), secondary (2°), or tertiary (3°).

- | | | |
|-----------------|--|--------------------|
| (a) butane | (b) isobutane | (c) 2-methylbutane |
| (d) cyclohexane | (e) norbornane (bicyclo[2.2.1]heptane) | |

4-13B Free-Radical Stabilities

Figure 4-7 shows the energy required (the bond-dissociation enthalpy) to form a free radical by breaking a bond between a hydrogen atom and a carbon atom. This energy is greatest for a methyl carbon, and it decreases for a primary carbon, a secondary carbon, and a tertiary carbon. The more highly substituted the carbon atom, the less energy required to form the free radical.

From the information in Figure 4-7, we conclude that free radicals are more stable when they are more highly substituted. The following free radicals are listed in increasing order of stability.



In the chlorination of propane, the secondary hydrogen atom is abstracted more often because the secondary radical and the transition state leading to it are lower in energy than the primary radical and its transition state. Using the bond-dissociation enthalpies in Table 4-2 (page 209), we can calculate ΔH° for each of the possible reaction steps. Abstraction of the secondary hydrogen is 13 kJ/mol (3 kcal/mol) more exothermic than abstraction of the primary hydrogen.

PROBLEM-SOLVING HINT

The first propagation step of chlorination is exothermic for all alkanes except methane. For methane, it is slightly endothermic, about +4 kJ/mol (+1 kcal/mol).

Formation of a methyl radical



Bond-dissociation enthalpy

$$\Delta H^\circ = 439 \text{ kJ (105 kcal)}$$

Formation of a primary (1°) radical



Formation of a secondary (2°) radical



Formation of a tertiary (3°) radical

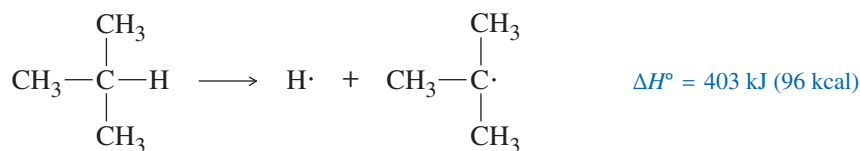


FIGURE 4-7 Enthalpy required to form a free radical. Bond-dissociation enthalpies show that more highly substituted free radicals are more stable than less highly substituted ones.

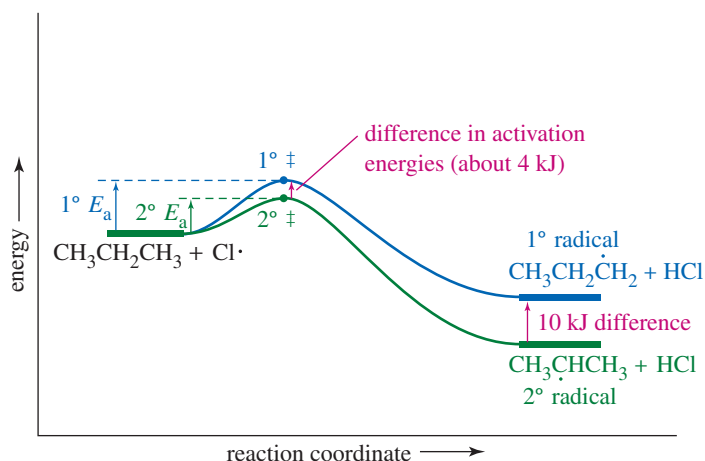
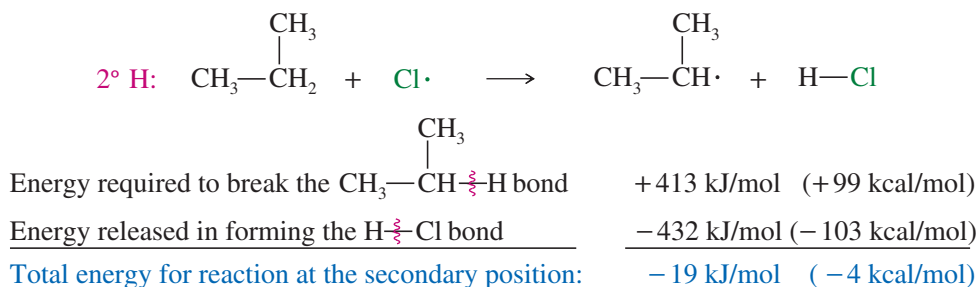
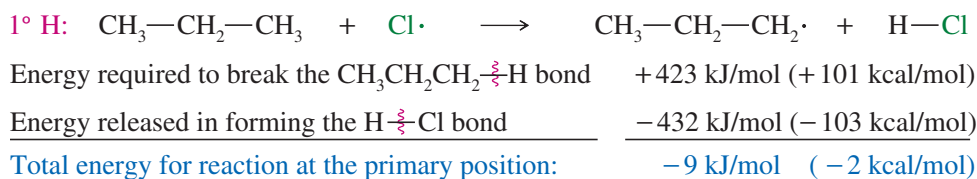


FIGURE 4-8 Reaction-energy diagram for the first propagation step in the chlorination of propane. Formation of the secondary radical has a lower activation energy than does formation of the primary radical.



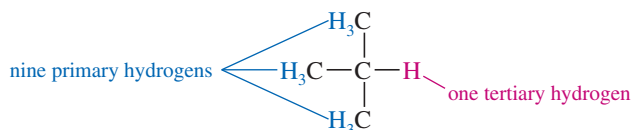
A reaction-energy diagram for this rate-limiting first propagation step appears in Figure 4-8. The activation energy to form the secondary radical is slightly lower, so the secondary radical is formed faster than the primary radical.

SOLVED PROBLEM 4-5

Tertiary hydrogen atoms react with $\text{Cl}\cdot$ about 5.5 times as fast as primary ones. Predict the product ratios for chlorination of isobutane.

SOLUTION

Isobutane has nine primary hydrogens and one tertiary hydrogen.



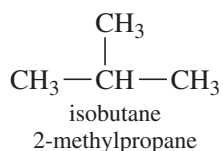
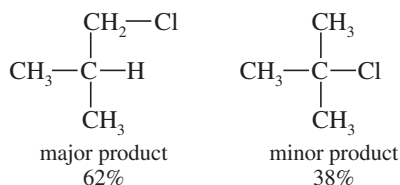
$$(9 \text{ primary hydrogens}) \times (\text{reactivity } 1.0) = 9.0 \text{ relative amount of reaction}$$

$$(1 \text{ tertiary hydrogen}) \times (\text{reactivity } 5.5) = 5.5 \text{ relative amount of reaction}$$

Even though the primary hydrogens are less reactive, there are so many of them that the primary product is the major product. The product ratio will be 9.0:5.5, or about 1.6:1.

$$\text{fraction of primary} = \frac{9.0}{9.0 + 5.5} = 62\%$$

$$\text{fraction of tertiary} = \frac{5.5}{9.0 + 5.5} = 38\%$$

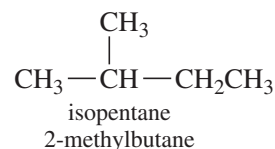


PROBLEM 4-20

Use the bond-dissociation enthalpies in Table 4-2 (page 209) to calculate the heats of reaction for the two possible first propagation steps in the chlorination of isobutane. Use this information to draw a reaction-energy diagram like Figure 4-8, comparing the activation energies for formation of the two radicals.

PROBLEM 4-21

Predict the ratios of products that result from chlorination of isopentane (2-methylbutane).

**PROBLEM 4-22**

- (a) When *n*-heptane burns in a gasoline engine, the combustion process takes place too quickly. The explosive detonation makes a noise called *knocking*. When 2,2,4-trimethylpentane (isooctane) is burned, combustion takes place in a slower, more controlled manner. Combustion is a free-radical chain reaction, and its rate depends on the reactivity of the free-radical intermediates. Explain why isooctane has less tendency to knock than does *n*-heptane.
- (b) Alkoxy radicals ($\text{R}-\text{O}\cdot$) are generally more stable than alkyl ($\text{R}\cdot$) radicals. Write an equation showing an alkyl free radical (from burning gasoline) abstracting a hydrogen atom from *tert*-butyl alcohol, $(\text{CH}_3)_3\text{COH}$. Explain why *tert*-butyl alcohol works as an antiknock additive for gasoline.
- (c) Use the information in Table 4-2 (page 209) to explain why toluene (PhCH_3) has a very high octane rating of 111. Write an equation to show how toluene reacts with an alkyl free radical to give a relatively stable radical.

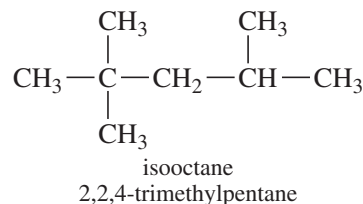
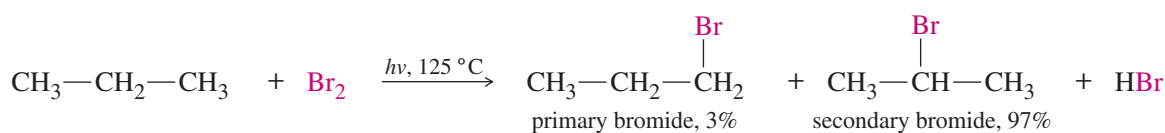
**4-13C Bromination of Propane**

Figure 4-9 shows the free-radical reaction of propane with bromine. Notice that this reaction is both heated to 125 °C and irradiated with light to achieve a moderate rate. The secondary bromide (2-bromopropane) is favored by a 97:3 product ratio. From this product ratio, we calculate that each of the two secondary hydrogens is 97 times as reactive as one of the six primary hydrogens.

The 97:1 reactivity ratio for bromination is much larger than the 4.5:1 ratio for chlorination. We say that bromination is more *selective* than chlorination because the major reaction is favored by a larger amount. To explain this enhanced selectivity, we must consider the transition states and activation energies for the rate-limiting step.

As with chlorination, the rate-limiting step in bromination is the first propagation step: abstraction of a hydrogen atom by a bromine radical. The energetics of the two possible hydrogen abstractions are shown on the next page. Compare these numbers with the energetics of the first propagation step of chlorination shown on page 221. The bond dissociation enthalpies are taken from Table 4-2 (page 209).

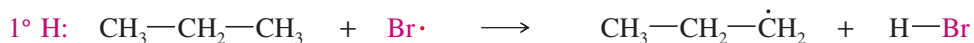
*Relative reactivity*

$$\text{six primary hydrogens} \quad \frac{3\%}{6} = 0.5\% \text{ per H}$$

$$\text{two secondary hydrogens} \quad \frac{97\%}{2} = 48.5\% \text{ per H}$$

$$\text{The } 2^\circ \text{ hydrogens are } \frac{48.5}{0.5} = 97 \text{ times as reactive as the } 1^\circ \text{ hydrogens.}$$

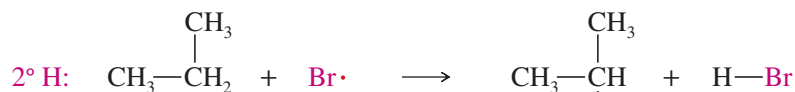
FIGURE 4-9 The free-radical reaction of propane with bromine. This 97:3 ratio of products shows that bromine abstracts a secondary hydrogen 97 times as rapidly as a primary hydrogen. Bromination (reactivity ratio 97:1) is much more selective than chlorination (reactivity ratio 4.5:1).



Energy required to break the $\text{CH}_3\text{CH}_2\text{CH}_2\text{—H}$ bond +423 kJ/mol (+101 kcal/mol)

Energy released in forming the H—Br bond −366 kJ/mol (−88 kcal/mol)

Total energy for reaction at the primary position: +57 kJ/mol (+13 kcal/mol)



Energy required to break the $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3\text{—CH—H} \end{array}$ bond +413 kJ/mol (+99 kcal/mol)

Energy released in forming the H—Br bond −366 kJ/mol (−88 kcal/mol)

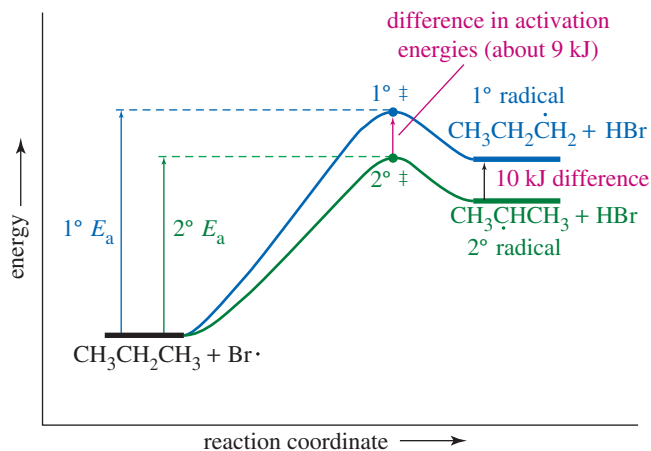
Total energy for reaction at the secondary position: +47 kJ/mol (+11 kcal/mol)

The energy differences between chlorination and bromination result from the difference in the bond-dissociation enthalpies of H—Cl (432 kJ) and H—Br (366 kJ). The HBr bond is weaker, and abstraction of a hydrogen atom by $\text{Br}\cdot$ is endothermic. This endothermic step explains why bromination is much slower than chlorination, but it still does not explain the enhanced selectivity observed with bromination.

Consider the reaction-energy diagram for the first propagation step in the bromination of propane (Figure 4-10). Although the difference in values of ΔH° between abstraction of a primary hydrogen and a secondary hydrogen is still 10 kJ/mol (2 kcal/mol), the energy diagram for bromination shows a much larger difference in activation energies for abstraction of the primary and secondary hydrogens than we saw for chlorination (Figure 4-8).

In bromination, the rate-limiting first propagation step is endothermic, and the energy maxima (corresponding to the activation energies) are closer to the products than to the reactants. A smooth graph (Figure 4-10) shows the activation energies nearly as far apart as the product energies. In chlorination, on the other hand, this first step is exothermic, and the energy maxima are closer to the reactants, which are the same and have the same energy for either route. The graph for chlorination (Figure 4-8) shows the activation energies separated by only a small fraction of the difference in product energies. This intuitive, graphic principle is formalized in **Hammond's postulate**.

FIGURE 4-10 Reaction-energy diagram for the first propagation step in the bromination of propane. The energy difference in the transition states is nearly as large as the energy difference in the products.



4-14 Hammond's Postulate

Hammond's postulate is a general statement summarizing the differences we have seen in comparing the chlorination and bromination of propane. We call it a postulate, rather than a rule, because we cannot prove it, yet it appears to be self-evident.

HAMMOND'S POSTULATE: Related species that are closer in energy are also closer in structure. The structure of a transition state resembles the structure of the closest stable species.

Figure 4-11 summarizes the energy diagrams for the first propagation steps in the bromination and chlorination of propane. Together, these energy diagrams explain the enhanced selectivity observed in bromination, and they show how Hammond's postulate applies to this reaction.

Two important differences are apparent in the reaction-energy diagrams for the first propagation steps of chlorination and bromination:

1. The first propagation step is endothermic for bromination but exothermic for chlorination.
2. The transition states forming the 1° and 2° radicals for the endothermic bromination have a larger energy difference than those for the exothermic chlorination, even though the energy difference of the products is the same (10 kJ, or 2 kcal) in both reactions.

Hammond's postulate tells us that the structures of the transition states in the endothermic bromination should resemble the products more than the reactants because they are closer in energy to the products. In the exothermic chlorination, the transition states are closer in energy to the reactants than the products; so we expect their structure to resemble the reactants more than the products. Figure 4-12 compares the transition states for bromination and chlorination.

In the product-like transition state for bromination, the C—H bond is nearly broken and the carbon atom has a great deal of radical character. The energy of this transition state reflects most of the energy difference of the radical products. In the reactant-like transition

George Simms Hammond

(1921–2005) was an American theoretical chemist who developed "Hammond's postulate." His pioneering work established the field of organic photochemistry, in addition to his research describing the geometric structures of transition states in organic chemical reactions. For his groundbreaking work, he was awarded the Priestley Medal in 1976.

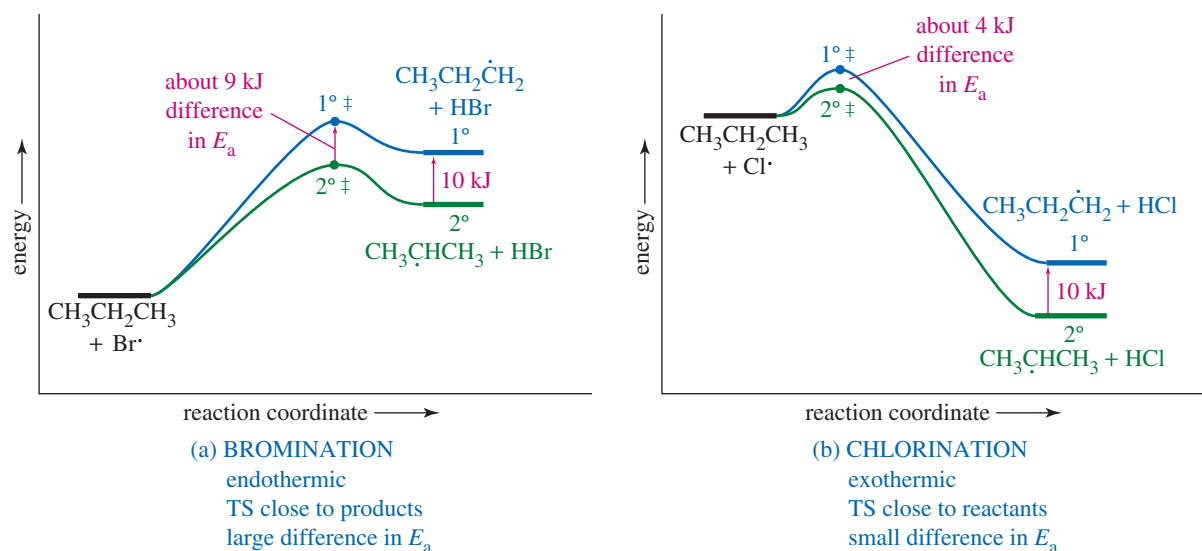
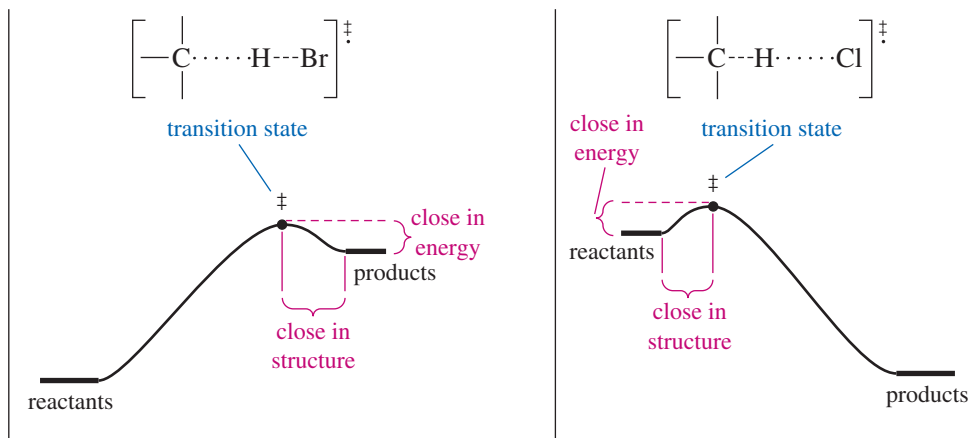


FIGURE 4-11 The energy diagrams for bromination and chlorination of propane. (a) In the endothermic bromination, the transition states are closer to the products (the radicals) in energy and in structure. The difference in the 1° and 2° activation energies is about 9 kJ (2.2 kcal), nearly the entire energy difference of the radicals.

(b) In the exothermic chlorination, the transition states are closer to the reactants in energy and in structure. The difference in activation energies for chlorination is about 4 kJ (1 kcal), less than half of the energy difference of the radicals.

FIGURE 4-12 Comparison of transition states for bromination and chlorination. In the endothermic bromination, the transition state resembles the products (the free radical and HBr). In the exothermic chlorination, the free radical has just begun to form in the transition state, so the transition state resembles the reactants.



state for chlorination, the C—H bond is just beginning to break, and the carbon atom has little radical character. This transition state reflects only a small part (less than half) of the energy difference of the radical products. Therefore, chlorination is less selective. We can state this conclusion as a general rule:

In an endothermic reaction, the transition state is closer to the products in energy and in structure. In an exothermic reaction, the transition state is closer to the reactants in energy and in structure.

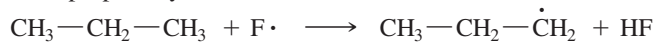
The transition state is always the point of highest energy on the energy diagram. Its structure resembles either the reactants or the products, whichever are higher in energy. In an endothermic reaction, the products are higher in energy, and the transition state is product-like. In an exothermic reaction, the reactants are higher in energy, and the transition state is reactant-like. Thus, the Hammond postulate helps us understand why exothermic processes tend to be less selective than similar endothermic processes.

PROBLEM-SOLVING HINT

Free-radical bromination is highly selective, chlorination is moderately selective, and fluorination is nearly nonselective.

PROBLEM 4-23

- (a) Compute the heats of reaction for abstraction of a primary hydrogen and a secondary hydrogen from propane by a fluorine radical.



- (b) How selective do you expect free-radical fluorination to be?
(c) What product distribution would you expect to obtain from the free-radical fluorination of propane?

PROBLEM-SOLVING STRATEGY Proposing Reaction Mechanisms

Throughout this course, we will propose mechanisms to explain reactions. We will discuss methods for dealing with different types of mechanisms as we encounter them. These techniques for dealing with a variety of mechanisms are collected in Appendix 3A. At this point, however, we focus on free-radical mechanisms like those in this chapter.

Free-Radical Reactions

General principles: Free-radical reactions generally proceed by chain-reaction mechanisms, using an initiator with an easily broken bond (such as chlorine, bromine, or a peroxide) to start the chain reaction. In drawing the mechanism, expect free-radical intermediates (especially highly substituted or resonance-stabilized intermediates). Watch for the most stable free radicals, and avoid any high-energy radicals such as hydrogen atoms.

1. Draw a step that breaks the weak bond in the initiator.

A free-radical reaction usually begins with an initiation step in which the initiator undergoes homolytic (free-radical) cleavage to give two radicals.

2. Draw a reaction of the initiator with one of the starting materials.

One of the initiator radicals reacts with one of the starting materials to give a free-radical version of the starting material. The initiator might abstract a hydrogen atom or add to a double bond, depending on what reaction leads to formation of the observed product. You might want to consider bond-dissociation enthalpies to see which reaction is energetically favored.

3. Draw a reaction of the free-radical version of the starting material with another starting-material molecule to form a bond needed in the product and to generate a new radical intermediate.

Check your intermediates to be sure that you have used the most stable radical intermediates. For a realistic chain reaction, no new initiation steps should be required; a radical should be regenerated in each propagation step.

4. Draw termination step(s).

The reaction ends with termination steps, which are side reactions rather than part of the product-forming mechanism. The reaction of any two free radicals to give a stable molecule is a termination step, as is a collision of a free radical with the reaction vessel.

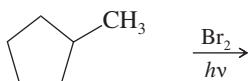
Before we illustrate this procedure, let's consider a few common mistakes. Avoiding these mistakes will help you to draw correct mechanisms throughout this course.

Common Mistakes to Avoid

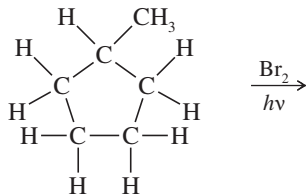
1. Do not use condensed or line-angle formulas for reaction sites. Draw all the bonds and all the substituents of each carbon atom affected throughout the mechanism. Three-bonded carbon atoms in intermediates are most likely to be radicals in the free-radical reactions we have studied. If you draw condensed formulas or line-angle formulas, you will likely misplace a hydrogen atom and show a reactive species on the wrong carbon.
2. Do not show more than one step occurring at once, unless they really do occur at once.

Sample Problem

Draw a mechanism for the reaction of methylcyclopentane with bromine under irradiation with light. Predict the major product.



In every mechanism problem, we first draw what we know, showing all the bonds and all the substituents of each carbon atom that may be affected throughout the mechanism.

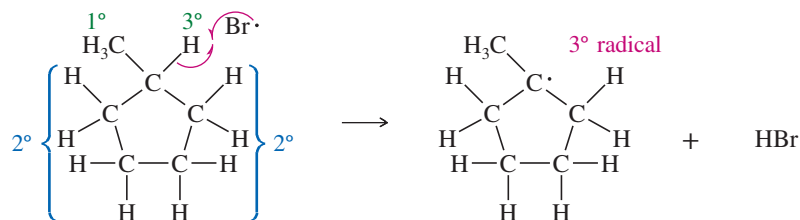
**1. Draw a step involving cleavage of the weak bond in the initiator.**

The use of light with bromine suggests a free-radical reaction, with light providing the energy for dissociation of Br₂. This homolytic cleavage initiates the chain reaction by generating two Br· radicals.

**2. Draw a reaction of the initiator with one of the starting materials.**

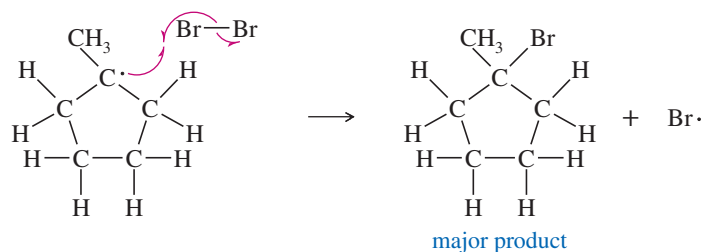
One of these initiator radicals should react with methylcyclopentane to give a free-radical version of methylcyclopentane. As we have seen, a bromine or chlorine radical can abstract a hydrogen atom from an alkane to generate an alkyl radical. The bromine radical is highly selective, and the most stable alkyl radical should result. Abstraction of the tertiary hydrogen atom gives a tertiary radical.

(continued)

First propagation step

3. Draw a reaction of the free-radical version of the starting material with another starting-material molecule to form a bond needed in the product and to generate a new radical intermediate.

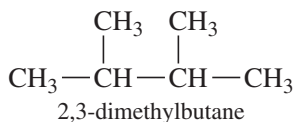
The alkyl radical should react with another starting-material molecule, in another propagation step, to generate a product and another radical. Reaction of the alkyl radical with Br_2 gives 1-bromo-1-methylcyclopentane (the major product) and another bromine radical to continue the chain.

Second propagation step

4. Draw termination step(s).

It is left to you to add some possible termination steps and summarize the mechanism developed here.

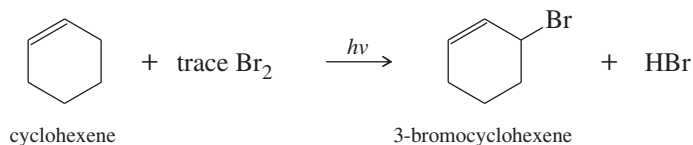
As practice in using a systematic approach to proposing mechanisms for free-radical reactions, work Problem 4-24 by going through the four steps just outlined.

**PROBLEM 4-24**

2,3-Dimethylbutane reacts with bromine in the presence of light to give a monobrominated product. Further reaction gives a good yield of a dibrominated product. Predict the structures of these products, and propose a mechanism for the formation of the monobrominated product.

PROBLEM 4-25

In the presence of a small amount of bromine, cyclohexene undergoes the following light-promoted reaction:



- Propose a mechanism for this reaction.
- Draw the structure of the rate-limiting transition state.
- Use Hammond's postulate to predict which intermediate most closely resembles this transition state.
- Explain why cyclohexene reacts with bromine much faster than cyclohexane, which must be heated to react.

4-15 Radical Inhibitors

We often want to prevent or retard free-radical reactions. For example, oxygen in the air oxidizes and spoils foods, solvents, and other compounds mostly by free-radical chain reactions. Chemical intermediates may decompose or polymerize by free-radical chain reactions. Even the cells in living systems are damaged by radical reactions, which can lead to aging, cancerous mutations, or cell death.

Radical inhibitors are often added to foods and chemicals to retard spoilage by radical chain reactions. Chain reactions depend on the individual steps being fast, so that each initiation step results in many molecules reacting, as in the reaction-energy diagram at the left in Figure 4-13. (Only the radicals are shown.)

The diagram at right in the figure shows how an inhibitor (I) can stop the chain by reacting with a radical intermediate in a fast, highly exothermic step to form an intermediate that is relatively stable. The next step in the chain becomes endothermic and very slow.

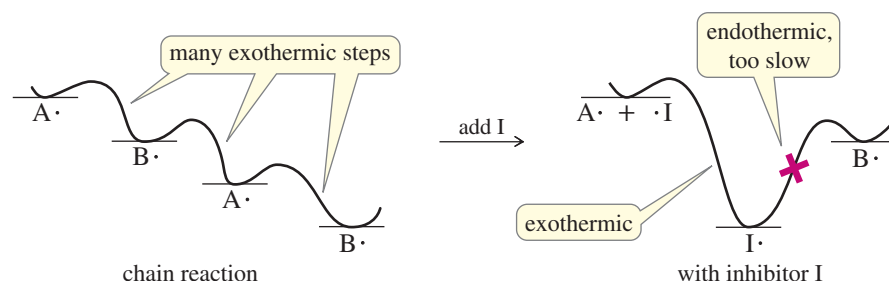
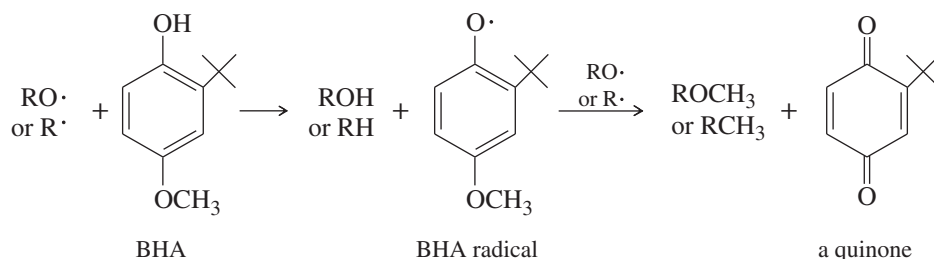


FIGURE 4-13 A radical chain reaction must be a series of exothermic (or mildly endothermic) steps because each step must go quickly, before a termination reaction occurs. An inhibitor reacts with a radical to give a relatively stable product that ends the chain, because it would require a strongly endothermic step for the chain reaction to continue.

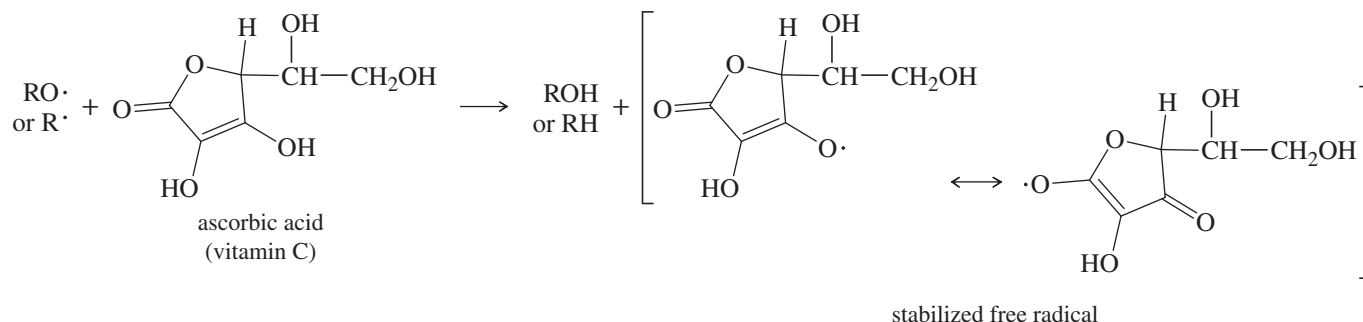
“Butylated hydroxyanisole” (BHA) is often added to foods as an antioxidant. It stops oxidation by reacting with radical intermediates to form a relatively stable free-radical intermediate (BHA radical). The BHA radical can react with a second free radical to form an even more stable quinone with all its electrons paired.

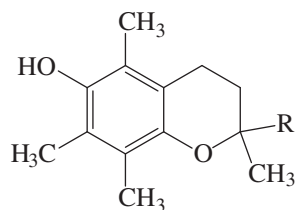


Application: Cancer Drugs

Some anti-cancer agents act by generating highly reactive hydroxyl radicals, which damage and degrade the DNA of the rapidly dividing tumor cells. As a result, the cells die and the tumor shrinks. One example of a radical generator is bleomycin, which is used for the treatment of testicular cancer.

Radical inhibitors also help to protect the cells of living systems. Like BHA, vitamin E is a *phenol* (an aromatic ring with an —OH group), and it is thought to react with radicals by losing the OH hydrogen atom as just shown for BHA. Ascorbic acid (vitamin C) is also thought to protect cells from free radicals, possibly by the following mechanism:





vitamin E
(R = alkyl chain)

PROBLEM 4-26

Draw resonance forms to show how the BHA radical is stabilized by delocalization of the radical electron over other atoms in the molecule.

PROBLEM 4-27

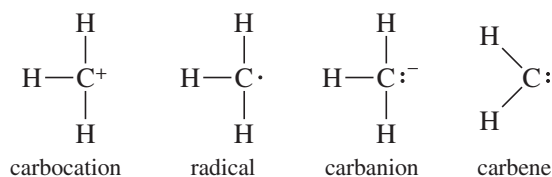
Write an equation for the reaction of vitamin E with an oxidizing radical ($\text{RO}\cdot$) to give ROH and a less reactive free radical.

4-16 Reactive Intermediates

The free radicals we have studied are one class of reactive intermediates. **Reactive intermediates** are short-lived species that are never present in high concentrations because they react as quickly as they are formed. In most cases, reactive intermediates are fragments of molecules (like free radicals), often having atoms with unusual numbers of bonds. Some of the common reactive intermediates contain carbon atoms with only two or three bonds, compared with carbon's four bonds in its stable compounds. Such species react quickly with a variety of compounds to give more stable products with tetravalent carbon atoms.

Although reactive intermediates are not stable compounds, they are important to our study of organic chemistry. Most reaction mechanisms involve reactive intermediates. If you are to understand these mechanisms and propose mechanisms of your own, you need to know how reactive intermediates are formed and how they are likely to react. In this chapter, we consider their structure and stability. In later chapters, we see how they are formed and ways they react to give stable compounds.

Species with trivalent (three-bonded) carbon are classified according to their charge, which depends on the number of nonbonding electrons. The *carbocations* have no nonbonding electrons and are positively charged. The *radicals* have one nonbonding electron and are neutral. The *carbanions* have a pair of nonbonding electrons and are negatively charged.



The most common intermediates with a divalent (two-bonded) carbon atom are the *carbenes*. A carbene has two nonbonding electrons on the divalent carbon atom, making it uncharged. Carbenes are not as common as carbocations, radicals, and carbanions.

4-16A Carbocations

A **carbocation** (also called a **carbonium ion** or a **carbenium ion**) is a species that contains a carbon atom bearing a positive charge. The positively charged carbon atom is bonded to three other atoms, and it has no nonbonding electrons, so it has only six electrons in its valence shell. It is sp^2 hybridized, with a planar structure and bond angles of about 120° . For example, the methyl cation ($^+\text{CH}_3$) is planar, with bond angles of exactly 120° . The unhybridized p orbital is vacant and lies perpendicular to the plane of the $\text{C}-\text{H}$ bonds (Figure 4-14). The structure of $^+\text{CH}_3$ is similar to the structure of BH_3 , discussed in Chapter 1.

With only six electrons in the positive carbon's valence shell, a carbocation is a powerful electrophile (Lewis acid), and it may react with any nucleophile it encounters. Like other strong acids, carbocations are unlikely to be found in basic solutions.

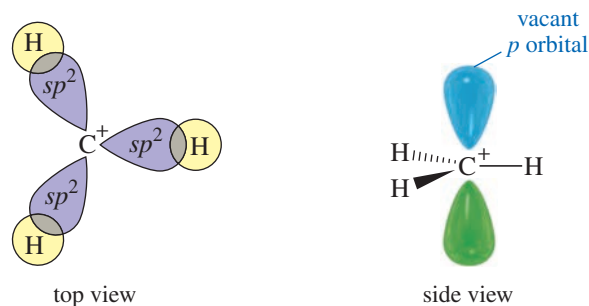


FIGURE 4-14 Orbital diagram of the methyl cation. The methyl cation is similar to BH_3 . The carbon atom is σ bonded to three hydrogen atoms by overlap of its sp^2 hybrid orbitals with the s orbitals of hydrogen. A vacant p orbital lies perpendicular to the plane of the three $\text{C}-\text{H}$ bonds.

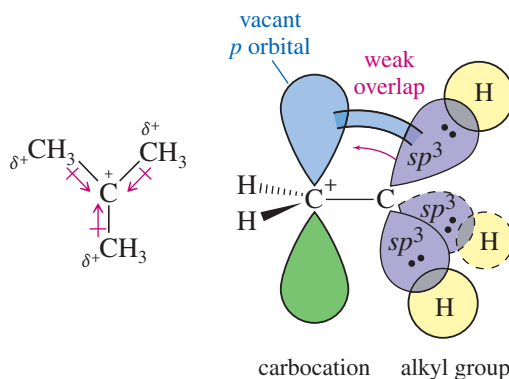


FIGURE 4-15 Effect of alkyl substituent on carbocation stability. A carbocation is stabilized by overlap of filled orbitals on an adjacent alkyl group with the vacant p orbital of the carbocation. Overlap between a σ bond and a p orbital is called *hyperconjugation*.

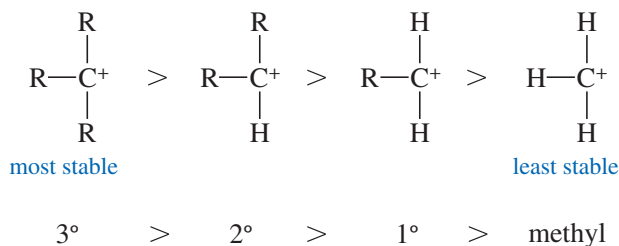
Carbocations are proposed as intermediates in many types of organic reactions, some of which we will encounter in Chapter 6.

Like free radicals, carbocations are *electron-deficient* species: They have fewer than eight electrons in the valence shell. Also like free radicals, carbocations are stabilized by alkyl substituents. An alkyl group stabilizes an electron-deficient carbocation in two ways: (1) through an inductive effect, and (2) through the partial overlap of filled orbitals with empty ones. The **inductive effect** is a donation of electron density through the sigma (σ) bonds of the molecule. The positively charged carbon atom withdraws some electron density from the polarizable alkyl groups bonded to it.

Alkyl substituents also have filled sp^3 orbitals that can overlap with the empty p orbital on the positively charged carbon atom, further stabilizing the carbocation (Figure 4-15). Even though the attached alkyl group rotates, one of its sigma bonds is always aligned with the empty p orbital on the carbocation. The pair of electrons in this σ bond spreads out into the empty p orbital, stabilizing the electron-deficient carbon atom. This type of overlap between a p orbital and a sigma bond is called *hyperconjugation*.

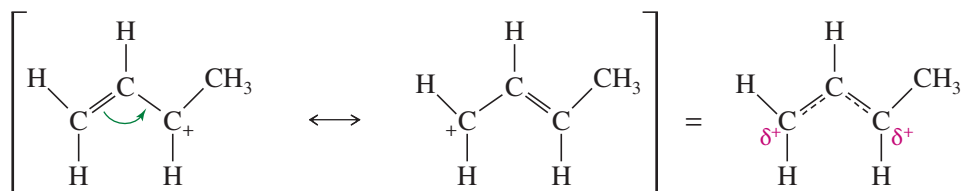
In general, more highly substituted carbocations are more stable.

Stability of carbocations

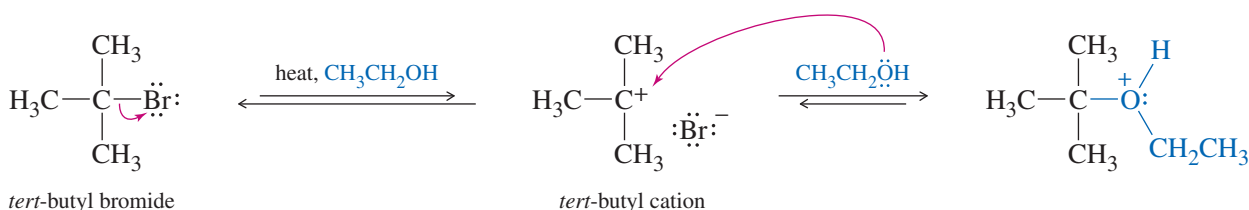


Unsaturated carbocations are also stabilized by **resonance stabilization**. If a pi (π) bond is adjacent to a carbocation, the filled p orbitals of the π bond will overlap

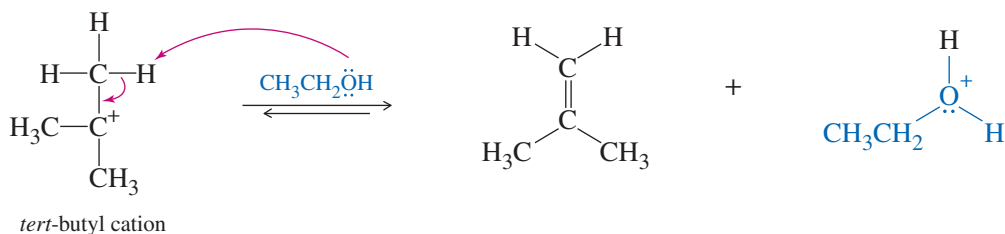
with the empty p orbital of the carbocation. The result is a delocalized ion, with the positive charge shared by two atoms. Resonance delocalization is particularly effective in stabilizing carbocations.



Carbocations are common intermediates in organic reactions. Highly substituted alkyl halides can ionize when they are heated in a polar solvent. The strongly electrophilic carbocation reacts with any available nucleophile, often the solvent.

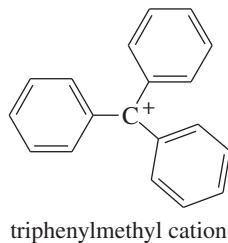


Carbocations are also strong proton acids. The *tert*-butyl cation shown above can also lose a proton to a weak base, often the solvent.



PROBLEM 4-28

The triphenylmethyl cation is so stable that some of its salts can be stored for months. Explain why this cation is so stable.



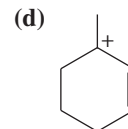
PROBLEM 4-29

Rank the following carbocations in decreasing order of stability. Classify each as primary, secondary, or tertiary.

(a) The isopentyl cation, $(\text{CH}_3)_2\text{CHCH}_2-\overset{+}{\text{CH}}_2$

(b) The 3-methyl-2-butyl cation, $\text{CH}_3-\overset{+}{\text{CH}}(\text{CH}_3)-\text{CH}(\text{CH}_3)_2$

(c) The 2-methyl-2-butyl cation, $\text{CH}_3-\overset{+}{\text{C}}(\text{CH}_3)\text{CH}_2\text{CH}_3$

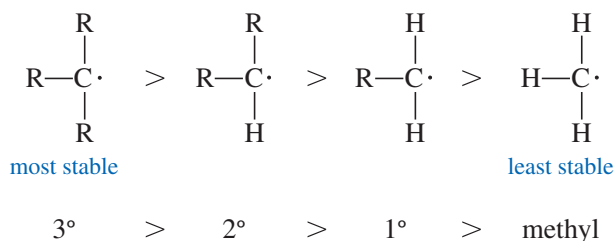


4-16B Free Radicals

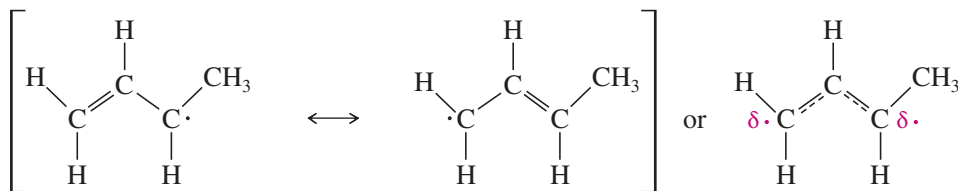
Like carbocations, **free radicals** are sp^2 hybridized and planar (or nearly planar). Unlike carbocations, however, the p orbital perpendicular to the plane of the C—H bonds of the radical is not empty; it contains the odd electron. Figure 4-16 shows the structure of the methyl radical.

Both radicals and carbocations are electron-deficient because they lack an octet around the carbon atom. Like carbocations, radicals are stabilized by the electron-donating effect of alkyl groups, making more highly substituted radicals more stable. This effect is confirmed by the bond-dissociation enthalpies shown in Figure 4-7: Less energy is required to break a C—H bond to form a more highly substituted radical.

Stability of radicals



Like carbocations, radicals can be stabilized by resonance. Overlap with the p orbitals of a π bond allows the odd electron to be delocalized over two carbon atoms. Resonance delocalization is particularly effective in stabilizing a radical.



PROBLEM 4-30

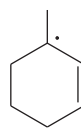
Rank the following radicals in decreasing order of stability. Classify each as primary, secondary, or tertiary.

(a) The isopentyl radical, $(\text{CH}_3)_2\text{CHCH}_2-\dot{\text{C}}\text{H}_2$

(b) The 3-methyl-2-butyl radical, $\text{CH}_3-\dot{\text{C}}\text{H}-\text{CH}(\text{CH}_3)_2$

(c) The 2-methyl-2-butyl radical, $\text{CH}_3-\dot{\text{C}}(\text{CH}_3)\text{CH}_2\text{CH}_3$

(d)



Application: Biochemistry

Vitamin B₁₂ is an essential dietary factor, and a deficiency results in anemia and neurological damage. The vitamin assists two different enzymes in the production and stabilization of methyl radicals. These methyl radicals are then used for the synthesis of important cellular components.

PROBLEM-SOLVING HINT

We describe atoms as being electron-deficient when they lack a full octet, whether or not they bear a formal charge. Recall that the boron atoms in BH₃ and BF₃ are uncharged, yet they are electron-deficient and are powerful electrophiles (Lewis acids). Similarly, carbocations (6 electrons, positive charge) and free radicals (7 electrons, uncharged) are both electron-deficient and highly reactive.

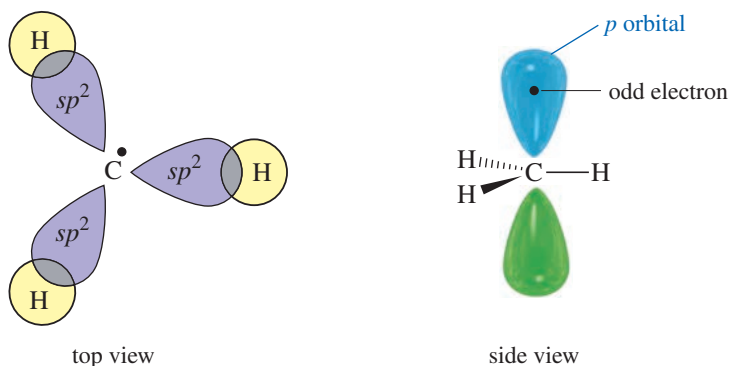
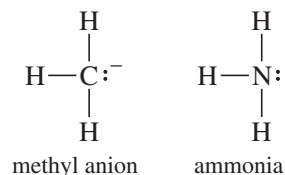
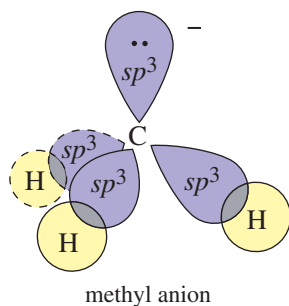


FIGURE 4-16 Orbital diagram of the methyl radical. The structure of the methyl radical is like that of the methyl cation (Figure 4-14), except there is an additional electron. The odd electron is in the p orbital perpendicular to the plane of the three C—H bonds.

4-16C Carbanions

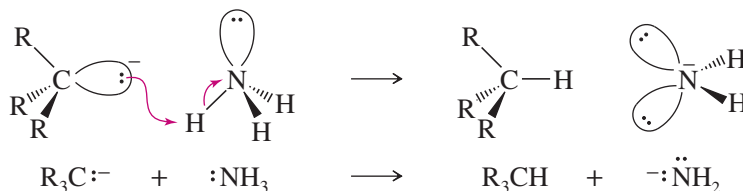
A **carbanion** has a trivalent carbon atom that bears a negative charge. There are eight electrons around the carbon atom (three bonds and one lone pair), so it is not electron-deficient; rather, it is electron-rich and a strong nucleophile (Lewis base). A carbanion has the same electronic structure as an amine. Compare the structures of a methyl carbanion and ammonia:



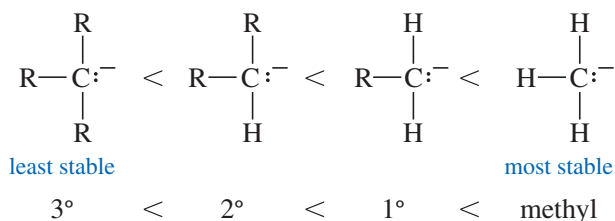
The hybridization and bond angles of a simple carbanion also resemble those of an amine. The carbon atom is sp^3 hybridized and tetrahedral. One of the tetrahedral positions is occupied by an unshared lone pair of electrons. Figure 4-17 compares the orbital structures and geometry of ammonia and the methyl anion.

Like amines, carbanions are nucleophilic and basic. A carbanion has a negative charge on its carbon atom, however, making it a more powerful base and a stronger nucleophile compared with an amine. For example, a carbanion is sufficiently basic to remove a proton from ammonia.

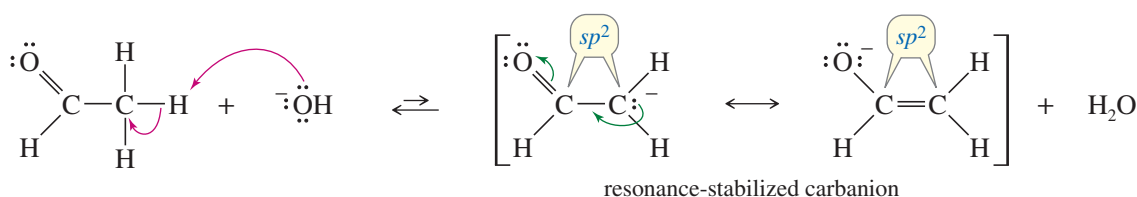
FIGURE 4-17 Comparison of orbital structures of the methyl anion and ammonia. Both the methyl anion and ammonia have an sp^3 hybridized central atom, with a nonbonding pair of electrons occupying one of the tetrahedral positions.



Like other strong bases, carbanions are unlikely to be found in acidic solutions. The stability order of carbanions reflects their high electron density. Alkyl groups and other electron-donating groups slightly destabilize a carbanion. The order of stability is usually the opposite of that for carbocations and free radicals.

Stability of carbanions

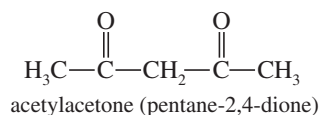
Carbanions that occur as intermediates in organic reactions are almost always stabilized by neighboring groups. They can be stabilized either by inductive effects or by resonance. For example, halogen atoms are electron-withdrawing, so they stabilize carbanions through the inductive withdrawal of electron density. Resonance also plays an important role in stabilizing carbanions. A carbonyl group ($C=O$) stabilizes an adjacent carbanion by overlap of its π bond with the nonbonding electrons of the carbanion. The negative charge is delocalized onto the electronegative oxygen atom of the carbonyl group.



This resonance-stabilized carbanion must be sp^2 hybridized and planar for effective delocalization of the negative charge onto oxygen (Section 1-17). Resonance-stabilized carbanions are the most common type of carbanions we will encounter in organic reactions.

PROBLEM 4-31

Acetylacetone (pentane-2,4-dione) reacts with sodium hydroxide to give water and the sodium salt of a carbanion. Write a complete structural formula for the carbanion, and use resonance forms to show the stabilization of the carbanion.

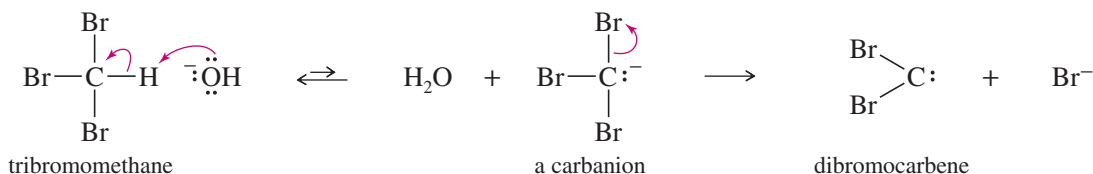


PROBLEM 4-32

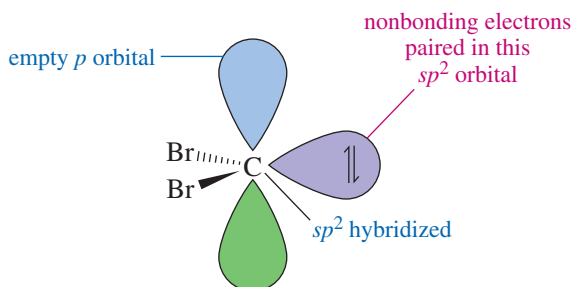
Propionitrile ($\text{:N}\equiv\text{C}-\text{CH}_2\text{CH}_3$) is deprotonated by very strong bases. Write resonance forms to show the stabilization of the carbanion that results.

4-16D Carbenes

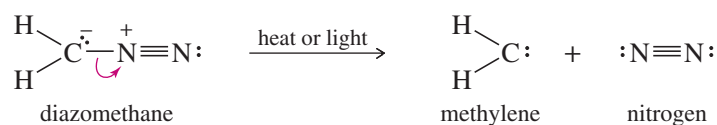
Carbenes are uncharged reactive intermediates containing a divalent carbon atom. The simplest carbene has the formula :CH_2 and is called *methylene*, just as a $-\text{CH}_2-$ group in a molecule is called a *methylene group*. One way of generating carbenes is to form a carbanion that can expel a halide ion. For example, a strong base can abstract a proton from tribromomethane (CHBr_3) to give an inductively stabilized carbanion. This carbanion expels bromide ion to give dibromocarbene.



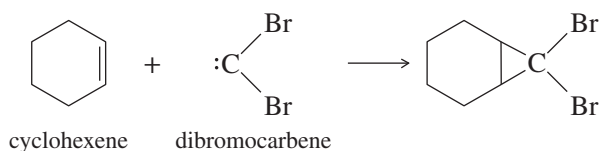
The carbon atom in dibromocarbene has only six electrons in its valence shell. It is sp^2 hybridized, with trigonal geometry. An unshared pair of electrons occupies one of the sp^2 hybrid orbitals, and there is an empty p orbital extending above and below the plane of the atoms. A carbene has both a lone pair of electrons and an empty p orbital, so it can react as a nucleophile or as an electrophile.



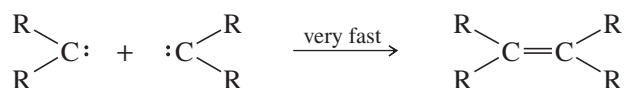
Methylene itself is formed when diazomethane (CH₂N₂) is heated or irradiated with light. The diazomethane molecule splits to form a stable nitrogen molecule and the very reactive carbene.



The most common synthetic reaction of carbenes is their addition to double bonds to form cyclopropane rings. For example, dibromocarbene adds to cyclohexene to give an interesting bicyclic compound.



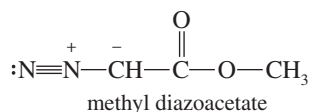
No simple carbenes have ever been purified or even made in a high concentration, because when two carbenes collide, they immediately dimerize (two of them bond together) to give an alkene.



Carbenes and carbenoids (carbene-like reagents) are useful both for the synthesis of other compounds and for the investigation of reaction mechanisms. The carbene intermediate is generated in the presence of its target compound, so that it can react immediately, and the concentration of the carbene is always low. Reactions using carbenes are discussed in Chapter 8.

PROBLEM 4-33

When it is strongly heated, methyl diazoacetate decomposes to give nitrogen gas and a carbene. Draw a Lewis structure of the carbene.



SUMMARY Reactive Intermediates

	Structure	Stability	Properties
carbocations	$\begin{array}{c} \\ -\text{C}^+ \\ \end{array}$	$3^\circ > 2^\circ > 1^\circ > ^+\text{CH}_3$	electrophilic strong acids
radicals	$\begin{array}{c} \\ -\text{C}\cdot \\ \end{array}$	$3^\circ > 2^\circ > 1^\circ > \cdot\text{CH}_3$	electron-deficient
carbanions	$\begin{array}{c} \\ -\text{C}^- \\ \end{array}$	$^-\text{CH}_3 > 1^\circ > 2^\circ > 3^\circ$	nucleophilic strong bases
carbenes	$\begin{array}{c} \diagup \\ \text{C} \\ \diagdown \end{array}:$		both nucleophilic and electrophilic

Essential Terms

activation energy (E_a) The energy difference between the reactants and the transition state; the minimum energy the reactants must have for the reaction to occur. (p. 213)

bond-dissociation enthalpy (BDE) The amount of enthalpy required to break a particular bond homolytically, to give radicals. (p. 207)



carbanion A strongly nucleophilic species with a negatively charged carbon atom having only three bonds. The carbon atom has a nonbonding pair of electrons. (p. 234)

carbene A highly reactive species with only two bonds to an uncharged carbon atom with a nonbonding pair of electrons. The simplest carbene is methylene, $:\text{CH}_2$. (p. 235)

carbocation (**carbonium ion, carbenium ion**) A strongly electrophilic species with a positively charged carbon atom having only three bonds. (p. 230)

catalyst A substance that increases the rate of a reaction (by lowering E_a) without being consumed in the reaction. (p. 215)

chain reaction A multistep reaction where a reactive intermediate formed in one step brings about a second step that generates the intermediate needed for the following step. (p. 199)

initiation step: The preliminary step in a chain reaction, where the reactive intermediate is first formed.

propagation steps: The steps in a chain reaction that are repeated over and over to form the product. The sum of the propagation steps should give the net reaction.

termination steps: Any steps where a reactive intermediate is consumed without another one being generated.

enthalpy (**heat content; H**) A measure of the heat energy in a system. In a reaction, the heat absorbed or evolved is called the *heat of reaction*, ΔH° . A decrease in enthalpy (negative ΔH°) is favorable for a reaction. (p. 205)

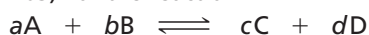
endothermic: Consuming heat (having a positive ΔH°).

exothermic: Giving off heat (having a negative ΔH°).

entropy (S) A measure of disorder or freedom of motion. An increase in entropy (positive ΔS°) is favorable for a reaction. (p. 205)

equilibrium A state of a system such that no more net change is taking place; the rate of the forward reaction equals the rate of the reverse reaction. (p. 203)

equilibrium constant A quantity calculated from the relative amounts of the products and reactants present at equilibrium. (p. 203) For the reaction



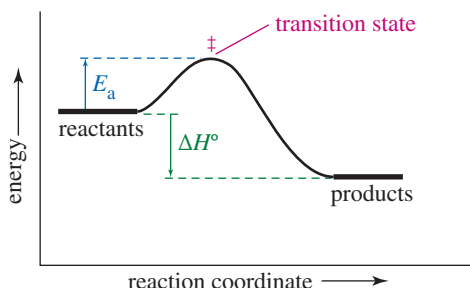
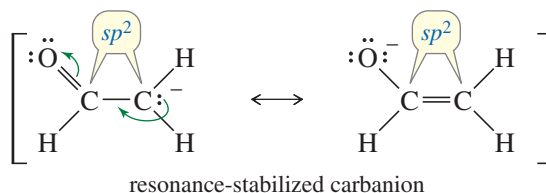
the equilibrium constant is

$$K_{\text{eq}} = \frac{[\text{C}]^c[\text{D}]^d}{[\text{A}]^a[\text{B}]^b}$$

free energy	(Gibbs free energy; G) A measure of a reaction's tendency to go in the direction written. A decrease in free energy (negative ΔG) is favorable for a reaction. (p. 204) Free-energy change is defined: $\Delta G = \Delta H - T\Delta S$
standard Gibbs free energy change:	(ΔG°) The free-energy change corresponding to reactants and products in their standard states (pure substances in their most stable states) at 25 °C and 1 atm pressure. ΔG° is related to K_{eq} by $K_{eq} = e^{-\Delta G^\circ/RT}$ (p. 204)
endergonic:	Having a positive ΔG° (unfavorable).
exergonic:	Having a negative ΔG° (favorable).
halogenation	The reaction of a halogen (X_2) or halogen-containing reagent that incorporates one or more halogen atoms into a molecule. Free-radical halogenation of alkanes is an important industrial synthesis, but it is rarely used in a laboratory setting. We study the reaction primarily because it serves as an uncomplicated example for studying its thermodynamics and kinetics. (pp. 197, 201) $R-H \xrightarrow[\text{heat or light}]{X_2} R-X + HX$
Hammond's postulate	Related species (on a reaction-energy diagram) that are closer in energy are also closer in structure. In an exothermic reaction, the transition state is closer to the reactants in energy and in structure. In an endothermic reaction, the transition state is closer to the products in energy and in structure. (p. 224)
heterolytic cleavage	(ionic cleavage) The breaking of a bond in such a way that one of the atoms retains both of the bond's electrons. A heterolytic cleavage forms two ions. (p. 208) $A:B \longrightarrow A:^- + B^+$
homolytic cleavage	(radical cleavage) The breaking of a bond in such a way that each atom retains one of the bond's two electrons. A homolytic cleavage produces two radicals. (p. 208) $A:B \longrightarrow A\cdot + \cdot B$
inductive effect	A donation (or withdrawal) of electron density through sigma bonds. (p. 231)
intermediate	A molecule or a fragment of a molecule that is formed in a reaction and exists for a finite length of time before it reacts in the next step. An intermediate corresponds to a relative minimum (a low point) in the reaction-energy diagram. (pp. 215, 221)
reactive intermediate:	A short-lived species that is never present in high concentration because it reacts as quickly as it is formed. (pp. 199, 200)
kinetics	The study of reaction rates. (pp. 198, 211)
mechanism	The step-by-step pathway from reactants to products, showing which bonds break and which bonds form in what order. The mechanism should include the structures of all intermediates and curved arrows to show the movement of electrons. (p. 198)
potential-energy diagram	See reaction-energy diagram . (p. 214)
radical	(free radical) A highly reactive species in which one of the atoms has an odd number of electrons. Most commonly, a radical contains a carbon atom with three bonds and an "odd" (unpaired) electron. (p. 200)
radical inhibitor	A compound added to prevent the propagation of free-radical chain reactions. In most cases, the inhibitor reacts to form a radical that is too stable to propagate the chain. (p. 229)
rate equation	(rate law) The relationship between the concentrations of the reagents and the observed reaction rate. (p. 211) A general rate law for the reaction $A + B \longrightarrow C + D$ is $\text{rate} = k_r[A]^a[B]^b$
order:	(kinetic order) The power of a concentration term in the rate equation. The preceding rate equation is a th order in $[A]$, b th order in $[B]$, and $(a + b)$ th order overall.
rate constant:	The proportionality constant k_r in the rate equation.
rate-limiting step	(rate-determining step) The slowest step in a multistep sequence of reactions. In general, the rate-limiting step is the step with the highest-energy transition state. (p. 216)

rate of a reaction

The amount of product formed or reactant consumed per unit of time. (p. 211)

reaction-energy diagram**(potential-energy diagram)** A plot of potential-energy changes as the reactants are converted to products. The vertical axis is potential energy (usually free energy, but occasionally enthalpy). The horizontal axis is the **reaction coordinate**, a measure of the progress of the reaction. (p. 214)**resonance stabilization**Stabilization that takes place by delocalization of electrons in a π bonded system. Cations, radicals, and anions are often stabilized by resonance delocalization. (p. 231)**substitution**

A reaction in which one atom replaces another, usually as a substituent on a carbon atom. (p. 199)

termination reaction

A step that produces fewer reactive intermediates (usually free radicals) than it consumes. (p. 202)

thermodynamics

The study of the energy changes accompanying chemical transformations. Thermodynamics is generally concerned with systems at equilibrium. (pp. 198, 203)

transition state**(activated complex)** The state of highest energy between reactants and products. A relative maximum (high point) on the reaction-energy diagram. (p. 214)

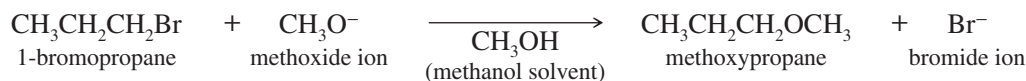
Essential Problem-Solving Skills in Chapter 4

Each skill is followed by problem numbers exemplifying that particular skill.

- | | | |
|----------|--|---|
| 1 | Propose a detailed mechanism for the free-radical halogenation of an alkane. | Problems 4-44, 46, 48, 51, 55, 56, and 58 |
| 2 | Predict the major halogenation products based on the stability of the intermediates and the selectivity of the halogenation. | Problems 4-43, 47, 48, and 49 |
| 3 | Draw a reaction-energy diagram for a mechanism, and point out the corresponding transition states, activation energies, intermediates, and rate-limiting steps. | Problems 4-35, 36, 37, and 38 |
| 4 | Use bond-dissociation enthalpies to calculate the enthalpy change for each step of a reaction and the overall enthalpy change for the reaction. | Problems 4-41, 51, 54, 55, 56, and 58 |
| 5 | Calculate free-energy changes from equilibrium constants, and calculate the position of an equilibrium from the free-energy changes. | Problems 4-53 and 58 |
| 6 | Determine the kinetic order of a reaction based on its rate equation. | Problems 4-34 and 39 |
| 7 | Use Hammond's postulate to predict whether a transition state will be reactant-like or product-like, and explain how this distinction affects the selectivity of a reaction. | Problems 4-50 and 57 |
| | <i>Problem-Solving Strategy: Proposing Reaction Mechanisms: Free-Radical Reactions</i> | Problems 4-44, 46, 48, 51, 55, 56, and 58 |
| 8 | Draw and describe the structures of carbocations, carbanions, free radicals, and carbenes and the structural features that stabilize them. Explain which are electrophilic and which are nucleophilic. | Problems 4-40, 42, 45, 46, 47, and 52 |

Study Problems

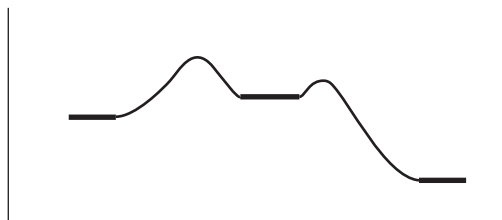
- 4-34 The following reaction is a common synthesis used in the organic chemistry laboratory course.



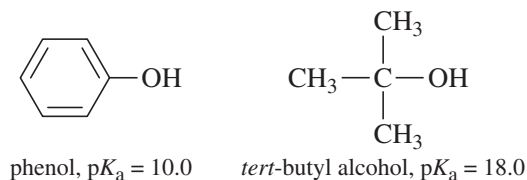
When we double the concentration of methoxide ion (CH_3O^-), we find that the reaction rate doubles. When we triple the concentration of 1-bromopropane, we find that the reaction rate triples.

- (a) What is the order of this reaction with respect to 1-bromopropane? What is the order with respect to methoxide ion? Write the rate equation for this reaction. What is the overall order?
- (b) One lab textbook recommends forming the sodium methoxide in methanol solvent, but before adding 1-bromopropane, it first distills off enough methanol to reduce the mixture to half of its original volume. What difference in rate will we see when we run the reaction (using the same amounts of reagents) in half the volume of solvent?

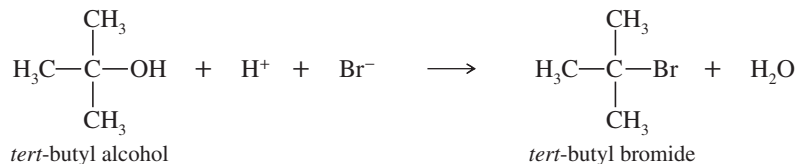
- 4-35 Consider the following reaction-energy diagram.



- (a) Label the reactants and the products. Label the activation energy for the first step and the second step.
- (b) Is the overall reaction endothermic or exothermic? What is the sign of ΔH° ?
- (c) Which points in the curve correspond to intermediates? Which correspond to transition states?
- (d) Label the transition state of the rate-limiting step. Does its structure resemble the reactants, the products, or an intermediate?
- 4-36 Draw a reaction-energy diagram for a one-step endothermic reaction. Label the parts that represent the reactants, products, transition state, activation energy, and heat of reaction.
- 4-37 Draw a reaction-energy diagram for a two-step endothermic reaction with a rate-limiting first step.
- 4-38 (a) Draw an approximate reaction-energy diagram for the acid–base reaction of phenol (see below) with 1-molar aqueous sodium hydroxide solution.
- (b) On the same diagram, draw an approximate reaction-energy diagram for the acid–base reaction of *tert*-butyl alcohol (see below) with 1-molar aqueous sodium hydroxide solution.



- 4-39 Treatment of *tert*-butyl alcohol with concentrated HBr gives *tert*-butyl bromide.



When the concentration of H^+ is doubled, the reaction rate doubles. When the concentration of *tert*-butyl alcohol is tripled, the reaction rate triples. When the bromide ion concentration is quadrupled, however, the reaction rate is unchanged. Write the rate equation for this reaction.

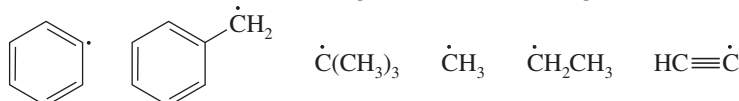
4-40 Label each hydrogen atom in the following compounds as primary (1°), secondary (2°), or tertiary (3°).



4-41 Use bond-dissociation enthalpies (Table 4-2, p. 203) to calculate values of ΔH° for the following reactions.



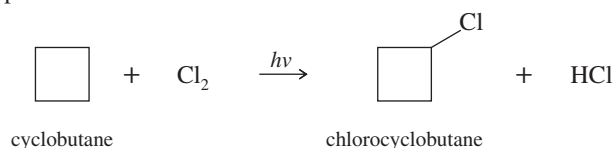
4-42 Use the information in Table 4-2 to rank the following radicals in decreasing order of stability.



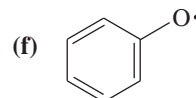
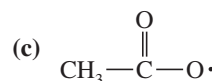
4-43 For each alkane,

- draw all the possible monochlorinated derivatives.
- determine whether free-radical chlorination would be a good way to make any of these monochlorinated derivatives. (Will the reaction give mostly one major product?)
- which monobrominated derivatives could you form in good yield by free-radical bromination?
 - cyclopentane
 - 2-methylpentane
 - methylcyclopentane
 - 2,2,3,3-tetramethylbutane

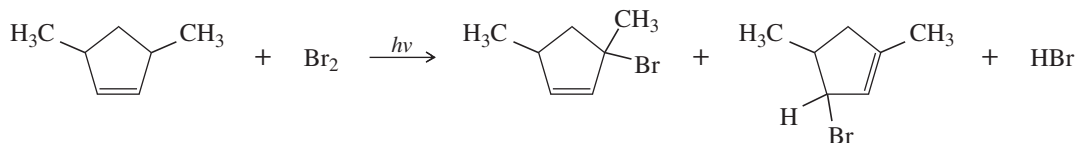
4-44 Write a mechanism for the light-initiated reaction of cyclobutane with chlorine to give chlorocyclobutane. Label the initiation and propagation steps.



4-45 Draw the important resonance forms of the following free radicals.

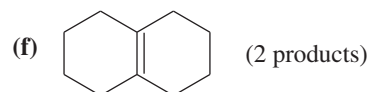


***4-46** In the presence of a small amount of bromine, the following light-promoted reaction has been observed.

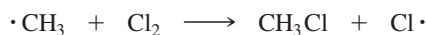


- (a) Write a mechanism for this reaction. Your mechanism should explain how both products are formed. (*Hint*: Notice which H atom has been lost in both products.)
- (b) Explain why only this one type of hydrogen atom has been replaced, in preference to any of the other hydrogen atoms in the starting material.

4-47 For each compound, predict the major product of free-radical bromination. Remember that bromination is highly selective, and only the most stable radical will be formed.



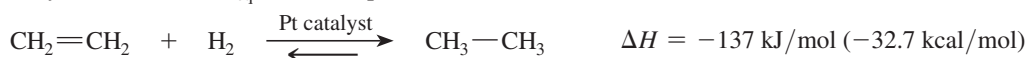
- 4-48 When exactly 1 mole of methane is mixed with exactly 1 mole of chlorine and light is shone on the mixture, a chlorination reaction occurs. The products are found to contain substantial amounts of di-, tri-, and tetrachloromethane, as well as unreacted methane.
- Explain how a mixture is formed from this stoichiometric mixture of reactants, and propose mechanisms for the formation of these compounds from chloromethane.
 - How would you run this reaction to get a good conversion of methane to CH_3Cl ? Of methane to CCl_4 ?
- 4-49 The chlorination of pentane gives a mixture of three monochlorinated products.
- Draw their structures.
 - Predict the ratios in which these monochlorination products will be formed, remembering that a chlorine atom abstracts a secondary hydrogen about 4.5 times as fast as it abstracts a primary hydrogen.
- 4-50 (a) Draw the structure of the transition state for the second propagation step in the chlorination of methane.



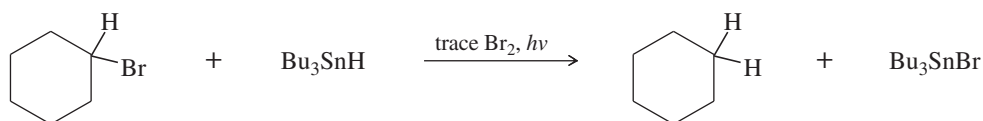
Show whether the transition state is product-like or reactant-like and which of the two partial bonds is stronger.

(b) Repeat for the second propagation step in the bromination of methane.

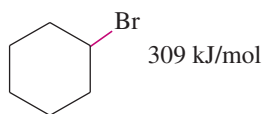
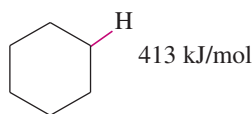
- 4-51 Peroxides are often added to free-radical reactions as initiators because the oxygen–oxygen bond cleaves homolytically rather easily. For example, the bond-dissociation enthalpy of the $\text{O}—\text{O}$ bond in hydrogen peroxide ($\text{H}—\text{O}—\text{O}—\text{H}$) is only 213 kJ/mol (51 kcal/mol). Give a mechanism for the hydrogen peroxide-initiated reaction of cyclopentane with chlorine. The BDE for $\text{HO}—\text{Cl}$ is 210 kJ/mol (50 kcal/mol).
- *4-52 When dichloromethane is treated with strong NaOH , an intermediate is generated that reacts like a carbene. Draw the structure of this reactive intermediate, and propose a mechanism for its formation.
- *4-53 When ethene is treated in a calorimeter with H_2 and a Pt catalyst, the heat of reaction is found to be -137 kJ/mol (-32.7 kcal/mol), and the reaction goes to completion. When the reaction takes place at 1400°K , the equilibrium is found to be evenly balanced, with $K_{\text{eq}} = 1$. Compute the value of ΔS for this reaction.



- *4-54 When a small amount of iodine is added to a mixture of chlorine and methane, it prevents chlorination from occurring. Therefore, iodine is a *free-radical inhibitor* for this reaction. Calculate ΔH° values for the possible reactions of iodine with species present in the chlorination of methane, and use these values to explain why iodine inhibits the reaction. (The $\text{I}—\text{Cl}$ bond-dissociation enthalpy is 211 kJ/mol or 50 kcal/mol.)
- *4-55 Tributyltin hydride (Bu_3SnH) is used synthetically to reduce alkyl halides, replacing a halogen atom with hydrogen. Free-radical initiators promote this reaction, and free-radical inhibitors are known to slow or stop it. Your job is to develop a mechanism, using the following reaction as the example.



The following bond-dissociation enthalpies may be helpful:



$\text{Br}—\text{Br}$	190 kJ/mol
$\text{H}—\text{Br}$	366 kJ/mol
$\text{Bu}_3\text{Sn}—\text{H}$	310 kJ/mol
$\text{Bu}_3\text{Sn}—\text{Br}$	552 kJ/mol

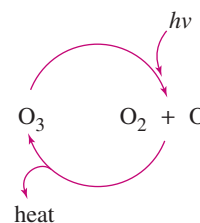
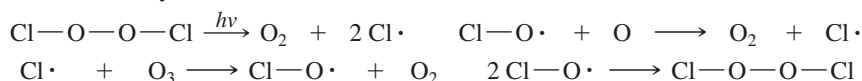
- Propose initiation and propagation steps to account for this reaction.
- Calculate values of ΔH for your proposed steps to show that they are energetically feasible. (*Hint:* A trace of Br_2 and light suggests it's there only as an initiator, to create $\text{Br}\cdot$ radicals. Then decide which atom can be abstracted most favorably from the starting materials by the $\text{Br}\cdot$ radical. That should complete the initiation. Finally, decide what energetically favored propagation steps will accomplish the reaction.)

- *4-56** When healthy, Earth's stratosphere contains a low concentration of ozone (O_3) that absorbs potentially harmful ultraviolet (UV) radiation by the cycle shown at right.

Chlorofluorocarbon refrigerants, such as Freon 12 (CF_2Cl_2), are stable in the lower atmosphere, but in the stratosphere they absorb high-energy UV radiation to generate chlorine radicals.



The presence of a small number of chlorine radicals appears to lower ozone concentrations dramatically. The following reactions are all known to be exothermic (except the one requiring light) and to have high rate constants. Propose two mechanisms to explain how a small number of chlorine radicals can destroy large numbers of ozone molecules. Which of the two mechanisms is more likely when the concentration of chlorine atoms is very small?



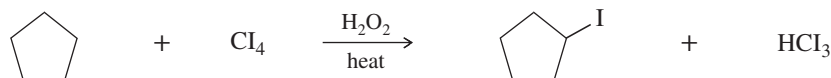
- *4-57** Deuterium (D) is the hydrogen isotope of mass number 2, with a proton and a neutron in its nucleus. The chemistry of deuterium is nearly identical to the chemistry of hydrogen, except that the $C-D$ bond is slightly stronger than the $C-H$ bond by 5.0 kJ/mol (1.2 kcal/mol). Reaction rates tend to be slower when a $C-D$ bond (as opposed to a $C-H$ bond) is broken in a rate-limiting step.

This effect, called a *kinetic isotope effect*, is clearly seen in the chlorination of methane. Methane undergoes free-radical chlorination 12 times as fast as tetradeuteriomethane (CD_4).



- Draw the transition state for the rate-limiting step of each of these reactions, showing how a bond to hydrogen or deuterium is being broken in this step.
- Monochlorination of deuterioethane (C_2H_5D) leads to a mixture containing 93% C_2H_4DCl and 7% C_2H_5Cl . Calculate the relative rates of abstraction per hydrogen and deuterium in the chlorination of deuterioethane.
- Consider the thermodynamics of the chlorination of methane and the chlorination of ethane, and use the Hammond postulate to explain why one of these reactions has a much larger isotope effect than the other.

- *4-58** Iodination of alkanes using iodine (I_2) is usually an unfavorable reaction. (See Problem 4-17, for example.) Tetraiodomethane (CI_4) can be used as the iodine source for iodination in the presence of a free-radical initiator such as hydrogen peroxide. Propose a mechanism (involving mildly exothermic propagation steps) for the following proposed reaction. Calculate the value of ΔH for each of the steps in your proposed mechanism.



The following bond-dissociation energies may be helpful:

